

CRF Complete Edition v17.17

Abstract & Contents Index

Pre-Work Audit Companion — Parts A, B, C, D, E, F, G, H, I, J, K, L
Navigation Index — not a derivation substitute (see How to Use)

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Abstract

This document is a single-file navigator for the CRF Complete Edition v17.16. It collects, in order, the volume-level abstract and the complete Part/Section header map for each of the twelve compiled parts (A–L), and additionally reproduces the individual ZC-paper abstracts for Parts C, D, E, F, I, J, K, and L where each integrated paper carries its own abstract.

v17.17 update (June 2026 — Part M added: the mass sector, end to end): Part M (NEW VOLUME) integrates ZC94–106 verbatim as a single continuous arc (Part LX). It carries the mass sector from framed problem to located residual. ZC94–98 reduce the charged-fermion spectrum to one running plus one $\pm d_s/2$ grading (spectral closure). ZC99 fixes the derivation boundary: the spectrum is parameter-free down to two constants, κ_R and L_0 . ZC100–102 trace those constants to an operation-group law — mass accumulates by addition (selecting $(\mathbb{Z}_{15}, +)$, order 15), force couples by a Clifford product (a distinct count) — with the closed form $\kappa_R = \ln(15)/(15\sqrt{d_s})$. GAP-ZC100-01 (the N-assignment) is CLOSED by ZC101/THM-ZC101-01, promoting CONJ-ZC100-01 HYP→DER; the residual GAP-ZC101-02 (0.23%) is left open. ZC103 builds the chain's first numerical simulator (zero free parameters; lepton gen2→3 confirmed to -0.2%). ZC104 excludes seven structural explanations of the first-rung (gen1→2) residual; ZC105 proves the mass sector does *not* instantiate the Part-L Q-pair law (THM-ZC105-01); ZC106 splits the residual into an inherited lepton piece (= GAP-MC-01, the ZC16 effect) and a genuinely new quark piece (THM-ZC106-01). Open at close of Part M: GAP-ZC101-02, GAP-ZC105-01 (split, both halves open), GAP-ZC95-01 / GAP-ZC98-01 (clarified, not closed). The arc stops at ZC106 by decision (a ZC107 quark first-rung paper would duplicate ZC104). Part M is v3.7-native; it inherits EQ-MC-03/GAP-MC-01/THM-MC-01 from Part C and the running bridge (GAP-ZC64-01→COR-ZC70-01) from Part I. Macro patch consolidated to `CRF_v17_17_macro_patch.tex` (= v17_16 + 25 net-new ZC94–106 macros; 263 total). No-Loss verified: 88 source atomic IDs identical; box counts identical except the added On-Ramp keybox and Open-Gaps box.

v17.16 update (June 2026 — Part L added: log-bridge unification + mind-layer structure): Part L (NEW VOLUME) integrates ZC88–93 verbatim across Parts LVII–LIX, in two clusters: Force-Layer Bridge (ZC88–89) and Mind-Layer Structure (ZC90–93). Continuous gap chain: GAP-ZC88-01 (which layer owns the bridge step) CLOSED by ZC89/THM-ZC89-01; GAP-ZC90-01 (are the two mind-steps distinct kinds) CLOSED by ZC91/THM-ZC91-01, which also RETIRES CONJ-ZC90-01 (irrationality-of-kind); GAP-ZC91-01 (forward-choice aggregation) PARTIALLY CLOSED by ZC92 (Barami functional, common rate); GAP-ZC92-01 (what gates the step kind) PARTIALLY CLOSED by ZC93/THM-ZC93-01 (gate-ratio). Open at close of Part L: GAP-ZC91-01 and GAP-ZC92-01 residuals. Part L is the first v3.7-native volume (Style Guide v3.7, UPG-30/31 prose-floor and header-contrast); it inherits the mind-layer pillars (DEF-MB03/04/05/06, EQ-MB01/02, P-MB04) from Part F and the floor/ Γ/ε structure from Part K. Notation:

PM-V17_16-EPSFAMILY continues the Part K ε -family override. Macro patch consolidated to `CRF_v17_16_macro_patch.tex` (single file; replaces the `v17_14 + v17_15` chain). No-Loss verified: 47 source atomic IDs identical; box counts identical except the added On-Ramp keybox and Connection-Map boxes.

v17.15 update (June 2026 — Part K added: ε -ontology closure arc + temporal bridge): Part K (NEW VOLUME) integrates ZC83–87 verbatim across Parts LIII–LVI. Four gap closures: GAP-ZC81-01 (Strong running bridge) CLOSED by ZC86/THM-ZC86-01; GAP-ZC83-01 (ε_{SC} ontology and closed form) CLOSED structurally by ZC85/FIND-ZC85-01 and in magnitude by ZC86/FIND-ZC86-01; GAP-ZC87-01 (joint-limit well-definedness) CLOSED by ZC87 v2; GAP-ZC76-01 structural sweep $\leftrightarrow$ dt bridge CLOSED by ZC87/THM-ZC87-01 (seconds-calibration remains externally blocked). New scars: PM-ZC83-01, PM-ZC84-SCOPE-01, PM-ZC85-01, PM-ZC86-01, PM-ZC87-01, PM-ZC87-02. The ε -floor reconciliation notice is extended to cover FIND-ZC83-01/FIND-ZC85-01 (same one-identity-three-layers structure). Notation: PM-V17_15-EPSFAMILY extends the Part J `\epsEM` override to the full ε family.

v17.14 update (June 2026 — ZC81/82 retraction integration + ε trap guard): COR-ZC81-02 and COR-ZC82-02 retracted (PM-ZC81-RETRACT-01, PM-ZC82-RETRACT-02); their Index entries updated. A boxed ε -floor reconciliation notice is added to the Quick-Reference Pack (with source-level `DO-NOT-RECONCILE` markers at each COR-ZC67-01 line) to stop the recurring “the floor values disagree” trap: COR-ZC67-01 is one identity evaluated on three input pairs (0.007550/0.007599/0.007605), all correct. This is an Index-level update over the *unchanged* v17.13 Part J file; “Part J v17.13” labels below refer to that file and are correct.

v17.13 update (May 2026 — Navigation Index upgrade): This release converts the Index from a Part-level navigator into a working *navigation index* suitable for a low-context (free-tier) session that does not load the full Complete Edition. Three additions: (1) per-paper abstracts for the entire ZC64–75 arc (Part I), matching the ZC14–40 format; (2) a front-loaded *Quick-Reference Pack* — T-canonical constants, the load-bearing exact equations, and a Master Open Gaps Registry spanning v17.6–v17.12; and (3) an explicit three-tier *missing-value protocol* in How to Use. The framing is deliberate: this Index is sufficient to know *what exists* and *what is open*, but it is **not** a substitute for the exact machinery of a source paper. When an exact form is not carried here, the rule is to fetch the one source paper or, failing that, ask the user — never to reconstruct a near-form (the ZC33 scar).

v17.9 update (May 2026): Part G added (NEW). Parts XXXV–XXXVII integrate ZC45–ZC49 verbatim in Part XXXV (ZC45 Universal Floor bridge) and Part XXXVI (ZC46–49 $\mathbb{Z}_{15}$ projection chain), with Cross-Part Connection Map v17.9 in Part XXXVII. Key closures: GAP-GJ-01 (ZC45 THM-ZC45-01), GAP-ZC46-01 (ZC47 COR-ZC47-01). Promotions: CONJ-ZC44-01 $\rightarrow$ THM-ZC46-01; CONJ-ZC47-01 $\rightarrow$ THM-ZC48-01; THM-ZC46-01 conditional $\rightarrow$ unconditional (ZC47). New scar tissue: PM-ZC45-01 (Wald residual), PM-ZC46-01, PM-ZC47-01, PM-ZC48-01 (probe scars), PM-ZC48-MACRO (`\Gcrf` convention conflict). PWA-6 codified (macro convention cross-check before embedding).

v17.8 update (May 2026): Part F added (NEW). Parts XXXI–XXXIV integrate ZC41–ZC44 verbatim in Part XXXI (physics/spectral cluster), the Mind/Barami layer in Part XXXII, the Axiom 11 Upgrade in Part XXXIII, and the Cross-Part Connection Map v17.8 in Part XXXIV. Key closures: GAP-ZC35-01 (ZC43, retroactive), GAP-ZC40-01 (ZC41), GAP-ZC42-01 (ZC44 via THM-ZC44-01), K14 promoted to exact unconditional Theorem (ZC41), THM-ZC42-01 promoted to Theorem (ZC44). New scar tissue: PM-AX11-01 (Axiom 11 upgrade), PM-ZC44-01 (M5/M7 CRT infrastructure overlap), OBS-ZC44-01 (mechanism count = $\varphi(15) = 8$). PWA-5 codified (mechanism-count audit before promotion).

v17.7 update (May 2026): Part E added (NEW). Parts XXVII–XXX integrate ZC31–ZC40 verbatim. Part XXX: Cross-Part Connection Map (NEW SECTION TYPE).

Primary use: Pre-Work Audit — before beginning any new ZC task, scan this Index to determine (a) what foundational machinery already exists, (b) which ZC papers have addressed adjacent problems, and (c) the exact atomic-unit labels available for inheritance.

Do not modify this index directly. It is generated from the source files and should be regenerated whenever a new version of any Part is released.

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How to Use This Index

New reader? Start with the Front Matter (v3.6 Communication Layer)

As of Edition v17.15, the Edition opens with a **Front Matter** document (CRF_Edition_FrontMatter_v1) written for a reader with no CRF background (Tier 0): “What is CRF in one page”, the **Map-Not-Reality** honesty clause, and the **Master Notation Key** (every recurring symbol in one clause). Each Complete-Edition Part now also opens with a plain-language “If you are just arriving” On-Ramp. This Index and the boxes serve the practitioner and auditor (Tier 1/2); the Front Matter and On-Ramps serve the newcomer (Tier 0). Governance: Style Guide v3.6 Section 31, Upgrade Protocol v1.3 Section U.

Pre-Work Audit Protocol — Quick Reference

Step 1. Identify your task’s domain (geometry, gauge, mass, thermodynamics, mind layer, ...).

Step 2. Scan the Contents Maps for Parts A–D below. Locate the Part(s) and Section(s) most relevant.

Step 3. Read the corresponding ZC-paper abstract (Parts C and D) to confirm what was established and what gaps remain open.

Step 4. Cross-reference the atomic-unit registry in the relevant paper before introducing any new DEF/CONJ/FIND/THM.

Rule: Existing CRF objects take priority over new constructions. Combine existing pillars before inventing.

Scope of This Index — Read Before Relying On It

This Index is a **navigation / triage layer**. It is sufficient to: (a) prevent duplication (know an object already exists); (b) run a Pre-Work Audit existence scan without opening source papers; (c) see the gap / PM / discovery topology of the whole programme.

It is **not** sufficient, on its own, to *reuse exact prior machinery* in a new derivation. A one-line registry entry tells you a theorem exists; it does not always carry the exact statement, domain conditions, or proof. The load-bearing exact forms are collected in the Quick-Reference Pack below and flagged “exact form in Index: Y”. For anything not so flagged, the one-liner is a pointer, not the object.

Domain-boundary caution (PWA-7). A flat one-line description strips the domain. The physics-layer spectral gap λ_2 and the mind-layer λ_1^{asava} both read as “a spectral gap” when compressed to one line — they are *not* interchangeable. Re-confirm the domain in the source before any cross-domain inheritance.

Missing-Value Protocol — Three-Tier Fallback (v17.13)

When a research value or exact form is needed and may not be in this Index, resolve it in this order. **Never guess or reconstruct a near-form** — that is the precise origin of the ZC33 scar (approximating $K^*T_{\text{avg}}/F = e$ instead of using the exact identity).

Tier 1 — In Index. The value / exact equation is in the Quick-Reference Pack (constant table or load-bearing equation table, flagged “Y”). Use it directly; cite the source ID.

Tier 2 — Fetch one source paper. Not in the Index, but the session can open the source. Fetch the single source paper for the object. *Mandatory trigger:* if a new paper

inherits from anything **older than ZC64**, fetch that source — the Index carries only a one-liner for deep-inheritance objects (Parts A–F machinery: T_{avg} , the TLS floor, K14, Γ_{CRF}).

Tier 3 — Ask the user. Not in the Index *and* the source cannot be opened (free-tier, context full). **Ask the user for the exact value or form.** Do not proceed on an approximation. This is the safe failure mode for a low-context session and protects correctness over convenience.

Volume overview:

Master Navigation Table. Each *volume* is a file named “Part A”...“Part J”. Inside the volumes, the finer-grained running “

Part I

,

’ units are numbered with Roman numerals I–LII. **These two uses of the word “Part” are distinct:** “Part I” (the file) is the ZC64–75 volume, whereas “Part I” as a Roman numeral is the very first foundations part inside Part A. The table below maps every volume to its Roman-numeral range, its ZC papers, and its theme.

Volume	Roman parts	ZC papers	Theme	Version
Part A	I–VIII	—	Foundations, Dynamics, Gravity, Wave Structure	v17.16^{†‡}
Part B	IX–XIII	—	Algebraic Closure, Geometry, Spectral, TLS, Trilogy	v17.16^{†‡}
Part C	XIV–XXI	Z2–Z4	Methodology, Z-Programme, Higgs VEV	v17.16[†]
Part D	XXIII–XXVI + XIX	ZC20–30	Lie Bridge, α_{EM} , R_1 Layer, Force Hierarchy	v17.16[†]
Part E	XXVII–XXX	ZC31–40	Cluster Closures, TLS Layer, Connection Map	v17.16[†]
Part F	XXXI–XXXIV	ZC41–44	Cluster, Mind/Barami, Axiom 11 Upgrade	v17.16[†]
Part G	XXXV–XXXVII	ZC45–49	$\mathbb{Z}_{15}$ Projection Chain, Four-Axis Duality	v17.16[†]
Part H	XXXVIII–XLIII	ZC50–63	Wald Self-Consistency, ε_0 from Axioms, Wald Correction	v17.16[†]
Part I	XLIV–XLVIII	ZC64–75	Prediction Registry, K14 Exact, Electroweak Gap, Gravity Bridge, $\mathbb{Z}_{15}$ Uniqueness	v17.16[†]
Part J	XLIX–LII	ZC76–82	Cosmological Bridge, Coupling Symbol, W_c Floor, TLS Quadratic, Electroweak Completeness, Strong Floor, Cosmo-Ladder	v17.16[†]
Part K	LIII–LVI	ZC83–87	Floor-Lattice Ordering, Complete Sector Synthesis, Observed-Bridge Bracket, Strong Running Bridge, Temporal Continuum Bridge	v17.15
Part L	LVII–LIX	ZC88–93	Log-Bridge (multiplicative $\leftrightarrow$ additive unification), Layer-Specificity, Cross-Layer Quantum-Pair, Irrationality-of-Kind, Barami Functional, Gate-Ratio	v17.16[§]
Part M	LX	ZC94–106	Mass-Hierarchy Registry, Mass Running Bridge, Tower Obstruction & Center Law, Spectral Closure, Derivation Boundary, Accumulation Integral, Operation-Group Law, Closure Discriminant, Mass-Sector Simulator, First-Rung Exclusion & Split	v17.17[§]

Reading order is alphabetical by volume (A through L); the section bodies in this index follow the same order.

[†]**Parts C–J (v17.16)** carry the full v3.6 Communication Layer: Reader’s On-Ramp, Bridge Prose at every ZC embed seam, one `\physmeaning` line, and a Map-Not-Reality pointer, added to the v17.6 Parts C/D, v17.7 Part E, v17.8 Part F, v17.9 Part G, v17.12 Parts H/I, and v17.13 Part J. With the Part K pilot, all nine Parts C–K now carry the layer. All boxes, atomic IDs, proofs, and numerics are unchanged from their source versions (No-Loss; body atomic-ID multisets verified identical). Per-Part notes: Part H required one pagination-only `\clearpage`; Parts D and I had stale copy-paste headers corrected; Parts C, D, E, F had macro lines repointed from absent per-version patches to the recovered/consolidated set (Part F also keeping its inline `\providecommand` blocks); Parts C and D each preserve one labelled placeholder for a simulator figure that never existed as a real artifact (No-Silent-Deletion). Only the prose layer outside the boxes was added.

‡**Parts A and B (v17.16)** are the foundations narrative and the algebraic-core narrative. They are prose-led \part-level volumes with no verbatim ZC embed seams, so per Style Guide v3.6 §31.7 they carry the *minimal* Communication Layer — Reader’s On-Ramp and audience-tier declaration only, no Bridge Prose or \physmeaning (there is no seam to attach them to, and the Parts already read as continuous prose). Both are self-contained (own macros; no external patch). No-Loss verified: body atomic-ID multisets identical, box counts identical except the one added On-Ramp keybox.

§**Part L (v17.16)** is the first *v3.7-native* volume: its Communication Layer (Reader’s On-Ramp, six Bridge-Prose seams, Connection Map) was authored under Style Guide v3.7 (UPG-30 prose-floor, UPG-31 header-contrast), not retrofitted. The six ZC88–93 sources already carry their own v3.7 in-paper bridge prose, embedded verbatim. All embedded \labels are namespaced -zcNN (with refs rewritten) to remove cross-paper collisions; no atomic ID, box, proof, or numeric is otherwise altered (No-Loss; 47-ID multiset verified identical, 0 multiply-defined, 0 undefined references).

1 Quick-Reference Pack (v17.13 Part J)

What This Pack Is For

The three tables below are the high-reuse core: the constants needed every session, the exact equations that other papers inherit, and the full open-gaps map. Everything flagged “In Index: Y” is safe to use directly (Tier 1 of the Missing-Value Protocol). Anything not here — or carried only as a one-line pointer in the per-paper abstracts — follows Tier 2/3 (fetch source, or ask user).

T-Canonical Constants

All values at T-canonical gauge $d_s = 3$ (exact), $n_m = 5$ (exact), $n = \varphi(15) = 8$, $|\mathbb{Z}_{15}| = 15$.

Symbol	Value	Closed form / source
d_s	3 (exact)	spatial dimension, gauge fix
n_m	5 (exact)	matter modes, gauge fix
$n = \varphi(15)$	8	Key Identity $3d_s = 2^{d_s} + 1$ (ZC72 FIND-ZC72-03)
$ \mathbb{Z}_{15} $	15	group order $\mathbb{Z}_3 \times \mathbb{Z}_5$
K^*	0.11750310	$1 - e^{-1/n}$ (LEM-ZC54-01)
$x \equiv n\varepsilon_{\text{can}}$	0.0604	canonical instability argument
φ_{gold}	1.6180339887	golden ratio, $\varphi_{\text{gold}}^2 = \varphi_{\text{gold}} + 1$
$\sin^2 \theta_W$	0.23213	$1 - \ln 8 / \ln 15$ (PRED-ZC64-01)
$\cos^2 \theta_W$	0.76787	$\ln 8 / \ln 15$
Δ_{Γ} (spectral step)	0.255958	$\cos^2 \theta_W / d_s = \ln 2 / \ln 15$ (FIND-ZC73-01)
ε_{can} (TLS floor)	0.007550	spectral self-consistency fixed point (ZC71)
$\varepsilon_{\text{univ}}$	0.007599	bridge fixed point; exact form COR-ZC69-02
ε_{EM}	0.007494 (exact)	$\varepsilon_{\text{univ}}(C_{\text{EM}}/C_W)^{1/4}$ (ZC70)
ε_W	0.007703 (exact)	$\varepsilon_{\text{univ}}(C_W/C_{\text{EM}})^{1/4}$ (ZC70)
$\varepsilon_W/\varepsilon_{\text{EM}}$	1.02809	$\sqrt{(1+2x)/(1+x)}$ (THM-ZC67-01)
$\delta\alpha/\alpha$	0.001322	$(d_s/2) \varepsilon_{\text{EM}} K^*$ (LEM-ZC70-01)
$\delta(\sin^2 \theta_W)$	0.000906	$\varepsilon_W K^*$ (CONJ-ZC66-01)
$\alpha_{\text{EM}}^{R_0}$	1/137.22	$(n/15) \Gamma_{\text{CRF}}$ (PRED-ZC64-02)
$\alpha_{\text{grav}}/\alpha_{\text{EM}}$	1/8 = $1/n$	exact (FIND-ZC72-01)
Γ_{CRF}	$\ln 2/D_0$	$= G^{R_0}/c^2$; $\alpha_{\text{grav}}^{R_0} \mathbb{Z}_{15} = \Gamma_{\text{CRF}}$ (ZC75)

ε FLOOR VALUES — VERIFIED CONSISTENT — DO NOT “RECONCILE” (read before editing any ε value)

If you (human or AI) see the near-equal floor values 0.007550, 0.007599, 0.007605 appearing alongside **COR-ZC67-01** and conclude they contradict — **they do not**. Audited line-by-line across ZC66–87; this is the single most common trap in the corpus and has cost multiple sessions.

COR-ZC67-01 (and **FIND-ZC83-01**, **FIND-ZC85-01**) is **ONE exact identity**, $\sqrt{\varepsilon_{\text{EM}} \cdot \varepsilon_W} = \varepsilon_{\text{base}}$, evaluated on a *different input pair at each layer*: • ZC66/67 defined pair (ε_{can} base) $\rightarrow 0.007550 = \varepsilon_{\text{can}}$; • ZC70/81/82/83 defined pair ($\varepsilon_{\text{univ}}$ base) $\rightarrow 0.007599 = \varepsilon_{\text{univ}}$ (R_0 -layer GM, FIND-ZC83-01); • observed pair (R_{max} layer) $\rightarrow 0.007605 = \varepsilon_{\text{SC}}$ (ZC68; “0.007605 = COR-ZC67-01, identical”; FIND-ZC85-01 shows it is the GM of the two FIND-ZC65-02 bridge floors; +0.08% above $\varepsilon_{\text{univ}}$, the ZC71 two-layer separation, with magnitude released by GAP-ZC81-01 closure in ZC86). These three are *intentionally distinct* (different input pairs at different R -layers), not rounding drift and not a contradiction. ZC83–87 add the layer reading and change *no* value.

DIRECTIVE. Do not change any ε value, do not “reconcile” them, do not open an audit, do not edit COR-ZC67-01 lines or the ZC81/82 retraction scars. More broadly, minor display spreads in floor values across papers (e.g. ε_W shown as 0.007703 here vs 0.007705 in some papers) are rounding/precision differences of the *same* quantity — verified, do *not* “fix” them. If a conflict still seems present, you are reading one identity as if it were several values: re-read this box, not the papers. (ZC66–69 audited and correct; ZC81/82 correct.)

Load-Bearing Exact Equations

The forms most often inherited by downstream papers, carried verbatim so a derivation need not open the source for the *statement* (the *proof* still lives in the source). “In Index: Y” = exact form present here.

ID	Exact form	Source	In Idx
THM-ZC20-01	$\sin^2 \theta_W = 1 - \ln 8 / \ln 15$	ZC20	Y
THM-ZC29-01	$\alpha_i = \varphi(d_i) / 15 \cdot \Gamma_{\text{CRF}}$	ZC29	Y
DEF-ZC29-04	$R_{\text{destroy},i} = 1 - \ln \varphi(d_i) / \ln 15$	ZC31	Y
FIND-ZC31-01	$\Gamma_i = \mathcal{G}_{\text{coup}} / d_s - 1 + \ln \varphi(d_i) / \ln 15$	ZC31	Y
THM-ZC73-01	$\Gamma_k = G / d_s - 1 + k(\cos^2 \theta_W / d_s), \quad k = 0, 1, 2, 3$	ZC73	Y
FIND-ZC73-01	$\Delta_\Gamma = \cos^2 \theta_W / d_s = \ln 2 / \ln 15$	ZC73	Y
COR-ZC73-01	$\Gamma_{\text{EM}} - \Gamma_{\text{vac}} = \cos^2 \theta_W$	ZC73	Y
FIND-ZC73-03	$\sum_{d_i 15} \varphi(d_i) = 1+2+4+8 = 15$	ZC73	Y
THM-ZC41-01 (K14)	$F \varphi_{\text{gold}}^2 = 4$ (exact)	ZC69	Y
LEM-ZC60-01	$F = 4 \lambda_2(\mathbb{Z}_5)^2 / n_m$	ZC60	Y
LEM-ZC56-01	$\cos(2\pi/5) = 1/(2\varphi_{\text{gold}})$	ZC56	Y
LEM-ZC69-01	$\lambda_2(\mathbb{Z}_5)^2 = 5/\varphi_{\text{gold}}^2$	ZC69	Y
THM-ZC42-01	$\lambda_2 = (1+x)/[n_m(1+2x)]$	ZC42	Y
COR-ZC62-02	$T_{\text{avg}} = eF/K^* = 2nK^*/[(n-1)\varepsilon_0]$	ZC62	Y
LEM-ZC62-01	$e = (1-K^*)^{-n}$ (exact)	ZC62	Y
LEM-ZC63-01	$W_c = 2n^2 K^*(1+x)/[n_m(n-1)x(1+2x)]$	ZC63	Y
THM-ZC57-01	two-sector correction $(1+x)/(1+2x)$	ZC57	Y
THM-ZC67-01	$\varepsilon_W / \varepsilon_{\text{EM}} = \sqrt{(1+2x)/(1+x)}$	ZC67	Y
COR-ZC67-01	$\sqrt{\varepsilon_{\text{EM}} \varepsilon_W} = \varepsilon_{\text{can}}$	ZC67	Y
COR-ZC69-02	$\varepsilon_{\text{univ}} = n(K^*)^2 \varphi_{\text{gold}}^2 / [2(n-1)e]$	ZC69	Y
LEM-ZC68-01	$\varepsilon_{\text{can}}^{\text{SC } 2} = \frac{\delta\alpha/\alpha \cdot \delta(\sin^2 \theta_W)}{(d_s/2)(K^*)^2}$	ZC68	Y
FIND-ZC65-01	$\delta\alpha/\alpha = (d_s/2) \varepsilon_0 K^*$	ZC65	Y
THM-ZC65-01	$\alpha(R_{\text{max}}) = \alpha(R_0)[1 + (d_s/2)\varepsilon_0 K^*]$	ZC65	Y
LEM-ZC70-01	$\frac{\delta\alpha/\alpha}{\delta(\sin^2 \theta_W)} = \frac{d_s}{2} \sqrt{\frac{1+x}{1+2x}}$	ZC70	Y
THM-ZC72-01	$\alpha_{\text{grav}}^{R_0} = \Gamma_{\text{CRF}}/15 = \alpha_{\text{EM}}^{R_0}/n$	ZC72	Y
THM-ZC75-01	$\alpha_{\text{grav}}^{R_0} \mathbb{Z}_{15} = G^{R_0} = \ln 2 / D_0 = \Gamma_{\text{CRF}}$	ZC75	Y
FIND-ZC75-02	$\ln(\alpha_{\text{EM}}^{R_0} / \alpha_{\text{grav}}^{R_0}) = d_s \ln 2 = \ln n$	ZC75	Y
PRED-ZC64-02	$\alpha_{\text{EM}}^{R_0} = (n/15) \Gamma_{\text{CRF}}$	ZC64	Y
Key Identity	$3d_s = 2^{d_s} + 1 \quad (d_s=3 \Rightarrow n=8)$	ZC72	Y
LEM-ZC74-01	$\mathbb{Z}_{15}$ unique squarefree semiprime, φ -set $\{1, 2, 4, 8\}$	ZC74	Y
LEM-ZC76-02	$w_{\text{DE}} = \mathcal{G}/d_s - 1 = -0.7542$	ZC76	Y
THM-ZC76-01	$P(t_{\text{now}})/P(0) = \mathcal{G}/d_s$	ZC76	Y
THM-ZC77-01	$\mathcal{G} = d_s(\Gamma_{\text{CRF}}^{\text{EM}} + \sin^2 \theta_W)$	ZC77	Y
THM-ZC79-01	$\varepsilon_{\text{TLS}} = 0.007550$ (quadratic root; exact)	ZC79	Y
THM-ZC80-01	$\delta \rightarrow 7$ EW observables; zero free params (R_0)	ZC80	Y
THM-ZC81-01	$\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}} = 0.007494$	ZC81	Y
THM-ZC82-01	$w_{\text{DE}} + 3\Delta_\Gamma = \Gamma_{\text{CRF}}^{\text{EM}}$	ZC82	Y
FIND-ZC82-01	$\Delta_\Gamma = \log_{15} 2$ (ladder bit)	ZC82	Y

ID	Exact form	Source	In Idx
THM-ZC83-01	floor lattice $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$ forced	ZC83	Y
FIND-ZC83-01	$\sqrt{\varepsilon_{\text{EM}}\varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ (R_0 GM; exact)	ZC83	Y
THM-ZC84-01	eight-fold sector synthesis from δ alone	ZC84	Y
FIND-ZC85-01	$\varepsilon_{\text{SC}} = \sqrt{\varepsilon_0^\alpha \varepsilon_0^{\text{sw}}}$ (R_{max} GM; exact)	ZC85	Y
THM-ZC86-01	Strong runs as EM; confinement in Γ not α	ZC86	Y
THM-ZC87-01	continuum dt = dense δ -sweep tick	ZC87	Y
THM-ZC88-01	log-bridge: multiplicative GP $\leftrightarrow$ additive AP, one law	ZC88	Y
FIND-ZC88-01	bridge is structural, not metric (steps not proportional)	ZC88	Y
THM-ZC89-01	layer-specificity fixes the bridge step to its layer	ZC89	Y
THM-ZC91-01	visesa/nibbedha irrational-of-kind (distinct, retires CONJ-ZC90-01)	ZC91	Y
THM-ZC93-01	gate-ratio governs which step-kind a choice may take	ZC93	Y

Master Open Gaps Registry (v17.6 $\rightarrow$ v17.16)

Consolidated gap topology. “Opened” = paper that registered the gap; “Successor/Target” = the gap it became or the planned closing paper. Priority is shown only where the source paper states it (otherwise “—”). Gaps closed in v17.13 (Part J) are marked **Closed (J)**; gaps closed in v17.15 (Part K) are marked **Closed (K)**; gaps closed or partially closed in v17.16 (Part L) are marked **Closed (L)** / **Partial (L)**; gaps closed, split, or re-seen in v17.17 (Part M) are marked **Closed (M)** / **Split (M)** / **Re-seen (M)**.

Gap ID	Description	Opened	Status (v17.12)	Successor/Target
GAP-ZC63-01	TLS-spectral self-consistency (0.66% separation)	ZC63	Closed (COR-ZC71-02); separation structural/irreducible	—
GAP-ZC64-01	R-layer running bridge α_{EM} $R_0 \rightarrow R_{\text{max}}$	ZC64	Closed (COR-ZC70-01)	—
GAP-ZC65-01	Exact ε_{TLS} from axioms	ZC65	Closed (J) (ZC79/COR-ZC79-01)	—
GAP-ZC66-01	Algebraic origin of $\varepsilon_{\text{W}}/\varepsilon_{\text{EM}} = 1.0280$	ZC66	Closed Near-Formal (COR-ZC67-02)	—
GAP-ZC67-01	Exact ε_{can} from axioms	ZC67	Inherited $\rightarrow$ partially resolved (ZC68)	GAP-ZC68-01
GAP-ZC68-01	Exact $\varepsilon_{\text{univ}}$ (needs K14 exact)	ZC68	Closed (COR-ZC69-01; K14 exact THM-ZC69-01)	—
GAP-ZC69-01	Numerical verify exact $\varepsilon_{\text{univ}}$ both bridges	ZC69	Closed (COR-ZC70-02)	—
GAP-ZC70-01	Observational precision (CO-DATA/PDG) upgrade	ZC70	Open (external)	ZC76+
GAP-ZC71-01	Exact $\Delta\varepsilon$ formula to all orders	ZC71	Closed (J) (ZC77/COR-ZC77-02 + ZC78/COR-ZC78-01)	—

Gap ID	Description	Opened	Status (v17.12)	Successor/Target
GAP-ZC29-02	Gravity sector identification from vacuum	ZC29	Partially closed at R_0 (COR-ZC72-01)	GAP-ZC72-01
GAP-ZC72-01	Cosmological bridge $\alpha_{\text{grav}}^{R_0} \rightarrow G_N(t)$	ZC72	Partially closed algebraic (COR-ZC75-01)	GAP-ZC75-01
GAP-ZC75-01	Cosmological evolution $P(t_{\text{now}})$	ZC75	Closed (J) (ZC76/THM-ZC76-01)	—
CONJ-ZC73-01	$\mathbb{Z}_{15}$ uniqueness (φ -set $\{1, 2, 4, 8\}$)	ZC73	Closed (COR-ZC74-01; 3×5 exclusive)	—
GAP-MC-01	Residual 10–17% in lepton mass ratios	ZC16	Re-seen (M) : split by ZC106/THM-ZC106-01 into lepton piece (= this gap, +18%) and new quark piece	GAP-ZC105-01
GAP-ZC21-00	Continuous Lie group from discrete $\mathbb{Z}_{15}$, $N \rightarrow \infty$	ZC21	Open	ZC22-ext
GAP-ZC36-02	Route A vs Route C structural gap (0.21%)	ZC36	Open	—
GAP-ZC77-01	$F_{\text{Wc}}^{\text{norm}}$ closed form	ZC77	Closed (J) (ZC78/COR-ZC78-01)	—
GAP-ZC78-01	Radical form of x_{TLS}	ZC78	Closed (J) (ZC79/THM-ZC79-01)	—
GAP-ZC76-01	$\tau_{\text{sweep}} = t_{\text{Planck}}$	ZC76	Closed (K) structural (ZC87/THM-ZC87-01); seconds-calibration externally blocked	—
GAP-ZC81-01	Strong sector $R_0 \rightarrow R_{\text{max}}$ running bridge	ZC81	Closed (K) (ZC86/THM-ZC86-01; confinement in Γ not α)	—
GAP-ZC83-01	ε_{SC} ontology and closed form	ZC83	Closed (K) : ontology ZC85/FIND-ZC85-01; magnitude ZC86/FIND-ZC86-01	—
GAP-ZC87-01	Joint-limit well-definedness (τ_{sweep} fixed)	ZC87	Closed (K) in v2 (ZC87/THM-ZC87-01)	—
Strong floor	$\varepsilon_{\text{strong}}$ floor (open since ZC30)	ZC30/81	Closed (J) (ZC81/THM-ZC81-01)	—

Gap ID	Description	Opened	Status (v17.12)	Successor/Target
GAP-ZC88-01	Which physical layer owns the log-bridge step	ZC88	Closed (L) (ZC89/THM-ZC89-01; layer-specificity)	—
GAP-ZC90-01	Are the two mind-steps distinct kinds	ZC90	Closed (L) (ZC91/THM-ZC91-01; retires CONJ-ZC90-01)	—
GAP-ZC91-01	How forward choices aggregate into capacity	ZC91	Partial (L) (ZC92; Barami functional, common rate)	residual open
GAP-ZC92-01	What gates the step kind of a choice	ZC92	Partial (L) (ZC93/THM-ZC93-01; gate-ratio)	residual open
GAP-ZC100-01	N-assignment ($\mathbb{Z}_{15}$ for mass-running vs. Clifford count for force)	ZC100	Closed (M) (ZC101/THM-ZC101-01; promotes CONJ-ZC100-01 HYP→DER)	—
GAP-ZC101-02	0.23% residual on $\kappa_R = \ln(15)/(15\sqrt{d_s})$	ZC101	Open	—
GAP-ZC105-01	First-rung residual (split: lepton = GAP-MC-01; quark new)	ZC105	Split (M) (ZC106/THM-ZC106-01)	both halves open
GAP-ZC95-01	Quark common-scale running slope	ZC95	Open (clarified, not closed)	—
GAP-ZC98-01	Pin L_0, κ from structure	ZC98	Open (restated, sharpened)	—

2 Part A — Foundations through Wave Structure (Parts I–VIII)

Volume Abstract

Part A — CRF Complete Edition v17.5 (Parts I–VIII)

The Constrained Reality Framework (CRF) proposes that all of physical and experiential reality is the continuous outcome of a single irreducible process: the probabilistic resolution of latent informational possibility under constraint. The entire structure derives from one Indefinable Primitive — Distinction (δ) — declared, as Euclid declared his points and lines, as the ground of all possibility.

This Complete Edition (v12) integrates all results through Paper v5, Sessions 1–21, Track A, the Wave Structure session, the Deep Structure session, Node Lifecycle v3, P0-H empirical work (7 runs, 3 subjects), and all auxiliary documents. Five results from v10 stand: (1) the Born Rule as a theorem via Gleason’s Theorem; (2) the Schwarzschild metric derived from non-linear $\mathcal{R}$ -feedback; (3) the gravitational constant $G = \ln(2) \cdot c^2 / D_0$ with no free parameters; (4) the full Einstein field equations with $T_{\mu\nu}$ derived from $\mathcal{C}$ -gradients; (5) the cosmic co-variation $G(t) \propto \Lambda(t) \propto 1/P(t)$.

New in v12–v16: Three deep identities ($D_0 = 1 - n\varepsilon_0$; $\theta(K^*) = K^*$; $\beta = n\varepsilon_0 T_{\text{avg}}$); Law of Exchange $|\beta| = d_s / (2 \ln 2)$ (Instability #21 closed at 0.67%); Resolution Scale (R_0, R_1, R_{max}) derived; Kerr metric, Cosmic rate $|\dot{G}/G| = 0.737 H_0$, K35 identity, Mind-Layer Bridge. Algebraic closure (Part IX): $\beta = d_s \mathcal{G}$, Clifford $\text{Cl}(3,0)$, three-phase universe, θ -tower, matter split $n = 3 + 5$, K35–K38 cluster, $\mathcal{K}$ coupling derived. Geometry crystallisation, spectral modes H1–H3, Grand Synthesis (Parts X–XI). TLS architecture, phase-ordered identities, bracket derivation, honest audit (Part XII). Foundational Trilogy + Methodology Programme (Parts XIII–XIV).

Pre-Work Audit note: Part A contains the foundational machinery: T_{avg} , Jaynes identity $K^* T_{\text{avg}} / F = e$, λ_2 , β -chain, Resolution Scale ($R_0/R_1/R_{\text{max}}$). These objects must be checked before constructing any new thermodynamic or timing-based quantity in ZC papers.

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3 Part B — Algebraic Core through Foundational Trilogy (Parts IX–XIII)

Volume Abstract

Part B — CRF Complete Edition v17.5 (Parts IX–XIII)

This is Part B of the CRF Complete Edition v17.5, containing Parts IX–XIII (Algebraic Closure, Geometry Crystallisation, Spectral Structure, TLS Architecture, Foundational Trilogy). Continues from Part A; preserves all v17.4 content verbatim.

Pre-Work Audit note: Part B is the primary source for: T_{avg} derivation (Part XI), Jaynes $K^*T_{\text{avg}}/F = e$ identity (Part XI), $\lambda_2 = (n - 1)/T_{\text{avg}}$ (Part XI), β -chain three derivation paths (Part XII), R-layer DEF-RN02 (Part XII), K-cluster identities K14/K35–K38 (Parts IX, XI), TLS architecture and dual- d_s lock (Part XII), Foundational Trilogy $S_{\text{dep}}/S_{\text{ind}}/\text{Barami}$ (Part XIII), Noether $\mathbb{Z}_{15}$ workstream setup (Part XIII).

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Volume Abstract

Part C — CRF Complete Edition v17.6 (Parts XIV–XXI)

This is **Part C** of the CRF Complete Edition v17.5, the third of three sequential files comprising one continuous volume on the δ -mechanism. Parts A and B (separately compiled) cover the foundations (Parts I–VIII) and the algebraic core (Parts IX–XIII) respectively. Part C contains **Parts XIV–XIX**, presenting the methodology layer, the self-applying framework, the closure of the Z-Programme, and the master status that locks the framework’s notation and companion documents.

Preserved from v17.4: Part XIV (Methodology Programme), Part XV (Self-Applying Framework), Part XVI (Z2 Programme Closure).

New in v17.5: Part XVII (Z3+Z4 Programme Closure) — THM-Z301, THM-HG01, THM-Z4a, THM-Z4b- α/β , PRED-Z4M01. Part XVIII (Bridge: Spontaneous to Agentive) — DEF-LX01 (Lexical Necessity), PRED-Z401 ($H_{\text{emp}} = 3.901 \pm 0.001$ bits, gap 0.138% from $\log_2 15$), THM-TC01 (χ -map identification), THM-PPP-01/02 and COR-PPP-01 (S_{ind} as PPP fixed point). Part XIX (Master Status, Notation Lock, Companion Embeds) — Master Z-Programme Status Table v17.5, Conflict Resolution Log, Symbol Registry v1.0 (Appendix A), Reality Not Simulation v3 (Appendix B).

New in v17.6 (Part C extension): Part XX (PPP Direct, Generations, Mass from Constraint) — ZC14, ZC15, ZC16. Part XXI (Higgs VEV from the δ -Chain) — ZC17, ZC18-A, ZC18-B, ZC19.

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4.1 ZC14 — PPP from δ -Symmetry Directly (Part XX)

ZC14: Closing GAP-PPP-01

GAP-PPP-01 (TASK-ZC13) asked whether the Possibility Preservation Principle (PPP) can be derived directly from the CRF δ -chain axioms, without passing through the assembled Z-Programme results. ZC13 identified a candidate derivation chain $\delta \rightarrow d_s = 3 \rightarrow \gcd(3, 5) = 1 \rightarrow \text{irreducibility} \rightarrow \text{PPP}$ but left the gap open, observing that the chain required treating the Key Identity as a first-principles result and that no axiom *explicitly* declared “ δ has no preferred mode.”

This paper closes GAP-PPP-01 completely. We show that the required property — δ -Symmetry: δ commutes with all permutations of the modes of $\mathcal{P}$ — is not a new axiom but a *corollary of Axiom 0 and Axiom 6 already in the CRF system*.

From δ -Symmetry, the Maximum Entropy principle (Jaynes 1957) immediately forces the δ -prior over $\mathcal{P}$ to be uniform. Combined with THM-ZC11-01 (irreducibility of the χ -map Markov chain on $\widehat{\mathbb{Z}}_{15}$), this yields $\pi_{\text{stat}} = \text{uniform}$, hence $\mathcal{V}_P = 15$, hence PPP.

Main result. THM-PD-01: GAP-PPP-01 is CLOSED. PPP is a theorem of Axiom 0 and Axiom 6 alone, modulo the Z-Programme middle structure. No new axiom is added to CRF.

4.2 ZC15 — Three Generations from the Subgroup Lattice (Part XX)

ZC15: Generations from $\mathbb{Z}_{15}$

CONJ-Z405 (TASK-ZC10, ZC13) proposed that the three non-trivial subgroups of $\mathbb{Z}_{15}$ correspond to the three observed SM fermion generations. This paper separates CONJ-Z405 into two parts with different provability status and treats each precisely.

Part 1 (THM-GEN-01, CLOSED). The *count* of three generations is derivable from the CRF δ -chain alone, without any physics input beyond the Z-Programme. The derivation uses the number-theoretic ω -function: $\delta \xrightarrow{\text{Key Identity}} d_s = 3, n_m = 5 \xrightarrow{\text{CRT}} |\mathbb{Z}_{15}| = 15 = 3 \times 5 \xrightarrow{\omega(15)=2} 2^{\omega(15)} - 1 = 3$. New identity: $\phi(15) = \phi(3)\phi(5) = 8 = n = |\mathcal{P}|$ (COR-GEN-02).

Part 2 (GAP-GEN-01, REMAINS OPEN). Why each successive generation is heavier requires a dynamical $\mathcal{V}_P$ -to-mass map. GAP-GEN-01 records this precisely. Falsifiable prediction PRED-ZC15-01 derived.

4.3 ZC16 — Mass from Constraint Cost (Part XX)

ZC16: Closing GAP-GEN-01 via Axiom 3

GAP-GEN-01 (TASK-ZC15) asked for a derivation of the Possibility Volume-to-mass map $f : \mathcal{V}_P \mapsto m$ from within CRF, without introducing free parameters.

This paper provides that derivation via Axiom 3. When a generation occupies subgroup $H \leq \widehat{\mathbb{Z}}_{15}$ with $\mathcal{V}_P = |H|$, each Resolution event pays a KL divergence cost $D_{\text{KL}}(\text{uniform}_H \| \text{uniform}_{\mathbb{Z}_{15}}) = \log(15/|H|) =: \log R$. This cost is paid independently by each of the $n_m = 5$ matter modes. Result: $m_H \propto e^{M_H} = R^{n_m} = (15/|H|)^5$. Zero free parameters.

Verification: Gen 2: $3^5 = 243$ vs SM 207 (+17%); Gen 3: $5^5 = 3125$ vs SM 3477 (−10%). GAP-MC-01 documents the residual.
COR-MC-03: S_{ind} has zero accumulated constraint — the massless ground state of the framework.

4.4 ZC17 — Higgs VEV from the δ -Chain (Part XXI)

ZC17: Information Ladder and Higgs VEV

This paper reports a three-phase investigation of whether the CRF δ -chain can derive $v = 246.22$ GeV without external input.

Phase 1 (ZC17-Prelim): Three open gaps identified. GAP-ZC17-02: EW gauge group $SU(2)_L \times U(1)_Y$ absent from δ -chain. GAP-ZC17-03: hierarchy $v/m_{\text{Pl}} \sim 10^{-17}$ unexplained.

Phase 3 (ZC17-Verify): Positive result is the **Information Ladder** (CONJ-ZC17-03): the hierarchy arises from $n = 8$ multiplicative suppressions of ε_0 . $n \cdot \ln(1/\varepsilon_0) \approx \ln(m_{\text{Pl}}/v)$ at 1.68% (OBS-ZC17-02). The target $\varepsilon_{\text{eff}} = (v_{\text{SM}}/m_{\text{Pl}})^{1/n} = 0.008186$ lies +8.4% above canonical ε_0 . Registered as open problem for ZC18. All failures registered as Phase Misalignments.

4.5 ZC18-A — TLS ε_0 Self-Consistency Sweep (Part XXI)

ZC18 Track A: PRED-ZC18-01 Falsified

PRED-ZC18-01 conjectured that ε_{eff} emerges as the $N \rightarrow \infty$ self-consistent fixed point of ε_0 .

This paper reports a complete investigation. Four self-consistency equations (SC1–SC4) of the TLS are derived and solved. All four are incompatible with ε_{eff} : SC3 yields 0.007843 (WaveChain constraint $F \cdot \varphi^2 = 4$).

Core finding: $\varepsilon_0 = 0.00755$ is fixed by definition via $I_{\text{eff}} = 0.7007$, not by finite- N convergence. PRED-ZC18-01 is **falsified** (Phase Misalignment).

Structural bridge identified: $\varepsilon_0^{\text{canon}}$ and ε_{eff} bracket the WaveChain target $F \cdot \varphi^2 = 4$ from opposite sides (−0.133% and +0.158%). Primary clue for Route B (EW sector coefficient derivation).

4.6 ZC18-B — $\mathbb{Z}_{15}$ Sector Correction (Part XXI)

ZC18 Track B: THM-ZC18-B01

Track A established that the gap $\varepsilon_{\text{eff}}/\varepsilon_0 = 1.0838$ is structural. This paper derives the correction from first principles.

Key observation: $\varphi(|\mathbb{Z}_{15}|) = \varphi(15) = 8 = n$. Candidate theorem: $\varepsilon_{\text{eff}}^{(B)} = \varepsilon_0 \cdot (15/8)^{1/8}$, with $\varepsilon_{\text{eff}}^{(B)} = 0.008170$ lying −0.19% from exact target.

The corrected Information Ladder: $n \cdot \ln(1/\varepsilon_0) - \ln(15/8) \approx \ln(m_{\text{Pl}}/v_{\text{SM}})$ at +0.04%. Higgs VEV prediction improves from −47.5% (ZC17) to −1.54%. THM-ZC18-B01 promoted. CONJ-ZC18-B01 registered (residual −1.54% attributed to Weinberg angle; deferred to ZC19).

4.7 ZC19 — The Weinberg Angle Candidate (Part XXI)

ZC19: FIND-ZC19-A01 and CONJ-ZC19-01

ZC18-B established THM-ZC18-B01 and registered residual -1.54% as attributed to the Weinberg angle (GAP-ZC17-02).

Central result (FIND-ZC19-A01, Candidate): $\sin^2 \theta_W \stackrel{?}{=} 1 - \ln n / \ln |\mathbb{Z}_{15}| = 1 - \ln 8 / \ln 15 \approx 0.2321$, lying $+0.39\%$ from measured $\sin^2 \theta_W(M_Z) = 0.2312$. When applied as correction coefficient, VEV prediction becomes $v_{\text{ZC}} = 246.25 \text{ GeV}$ (gap $+0.01\%$ from v_{SM}). GAP-ZC17-03 effectively closed.

Formal derivation from DEF-Z205 (χ -map) registered as CONJ-ZC19-01 and deferred to ZC20 (Part D).

5 Part D — Lie Bridge through Force Hierarchy (Parts XXIII–XXVI + XIX)

Volume Abstract

Part D — CRF Complete Edition v17.6 (Parts XXIII–XXVI + XIX)

This is **Part D** of the CRF Complete Edition v17.6, the fourth of four sequential files comprising one continuous volume on the δ -mechanism.

Part D contains **Parts XXIII–XXVI** (the continuous-Lie and R_0/R_1 layer dynamics) and the relocated **Part XIX** (Master Status, Symbol Registry, Reality Not Simulation), expanded to reflect the v17.6 closures.

Part XXIII (Lie Bridge): ZC20 derives $\sin^2 \theta_W = 1 - \ln n / \ln |\mathbb{Z}_{15}|$ formally (THM-ZC20-01, closing GAP-ZC17-02 at mixing-angle level). ZC21-v2 introduces CRF Thermodynamic Lie Algebra with $\hbar_{\text{eff}} = 1/N$. ZC22-v2 verifies $(\mathbb{Z}_{15})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4$ with sector identification to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ (THM-ZC22-01).

Part XXIV (α_{EM} Chain): ZC23 establishes non-abelian χ_{NA} partition. ZC24 identifies $\alpha_{\text{EM}}^{R_0} = (8/15)\Gamma_{\text{CRF}}$ (gap +0.132%). ZC28-v3 closes Γ_{CRF} construction (THM-ZC28-01: $\Gamma_{\text{CRF}} = \mathcal{G}/d_s - \sin^2 \theta_W$).

Part XXV (R_1 Crossing Layer): ZC25-v2 propagator-weight identification. ZC26 back-calculates R_1 -layer cost target $T(R_1)_{\text{EM}} = 3.421 \times 10^{-5}$. ZC27 derives via DEF-RN02 + φ -cancellation (THM-ZC27-01: $\delta_{\text{DKL}}^{R_1} = (d_s/n_m)\varepsilon_0^2$, gap 0.025%).

Part XXVI (Force Hierarchy & Sector Crossing): ZC29 proves THM-ZC29-01: $\alpha_i^{R_0} = (\varphi(d_i)/15)\Gamma_{\text{CRF}}$. ZC30 closes GAP-ZC29-01 for Weak and Strong sectors with φ -universality (FIND-ZC30-01).

Part XIX (relocated, expanded): Master Z-Programme Status Table v17.6, Phase Misalignment Registry v17.6, Symbol Registry v1.0 (Appendix A), Reality Not Simulation v3 (Appendix B).

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5.1 ZC20 — Formal Derivation of $\sin^2 \theta_W$ (Part XXIII)

ZC20: THM-ZC20-01 — GAP-ZC17-02 Closed

TASK-ZC19 identified the candidate formula $\sin^2 \theta_W = 1 - \ln n / \ln |\mathbb{Z}_{15}| \approx 0.2321$ (+0.39% from measured) and registered CONJ-ZC19-01. This paper completes all three sub-tasks and closes CONJ-ZC19-01.

Central result (THM-ZC20-01): *The Weinberg angle is determined by the generator partition of the CRF charge group $\mathbb{Z}_{15}$: $\cos^2 \theta_W = \ln \varphi(|\mathbb{Z}_{15}|) / \ln |\mathbb{Z}_{15}| = \ln n / \ln |\mathbb{Z}_{15}|$, where $\varphi(|\mathbb{Z}_{15}|) = n$ is structural identity ID-B01.*

Derivation chain: $\delta \rightarrow \mathbb{Z}_{15} \rightarrow \varphi(15) = n \rightarrow \mathbb{G}_{\text{gen}} \rightarrow \cos^2 \theta_W = \ln n / \ln |\mathbb{Z}_{15}|$. GAP-ZC17-02 (EW gauge group from the δ -chain) is closed.

5.2 ZC21-v2 — The Thermodynamic Lie Bridge (Part XXIII)

ZC21: CONJ-ZC21-01 — Candidate

GAP-ZC21-00 (inherited): derivation of the continuous Lie group structure $\text{SU}(2)_L \times U(1)_Y$ from the discrete $\mathbb{Z}_{15}$ in the thermodynamic limit.

Central result (CONJ-ZC21-01, Candidate): $\lim_{N \rightarrow \infty} \mathfrak{g}_{\text{CRF}}(N) \cong \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$, where $\mathfrak{g}_{\text{CRF}}(N)$ is the CRF Thermodynamic Lie Algebra (DEF-ZC21-01) on $\mathbb{G}_{\text{gen}} \cong (\mathbb{Z}_{15})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4$ with deformation parameter $\hbar_{\text{eff}} = 1/N$. Sector identification: $\mathbb{Z}_4 \leftrightarrow \mathfrak{su}(2)_L$; $\mathbb{Z}_2 \leftrightarrow \mathfrak{u}(1)_Y$.

Supporting lemmas: LEM-ZC21-01 (complex doubling, $H_\sigma = 2$ bits); LEM-ZC21-02 (dimensional counting: $\dim_{\mathbb{R}}(\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y) = n/2$). Observer interpretation: FIND-

ZC21-O01 identifies $N \rightarrow \infty$ with S_{ind} at gauge-symmetry level.

5.3 ZC22-v2 — Lie Bridge Verification (Part XXIII)

ZC22: THM-ZC22-01 and Tensor Product Discovery

CONJ-ZC21-01 opened three sub-tasks for verification.

Sub-task (ii) confirmed (THM-ZC22-01): $(\mathbb{Z}_{15})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4$ with explicit generators $g = 2$ (order 4) and $g = 11$ (order 2), and CRT sector identification $(\mathbb{Z}_5)^\times \cong \mathbb{Z}_4 \leftrightarrow \mathfrak{su}(2)_L$; $(\mathbb{Z}_3)^\times \cong \mathbb{Z}_2 \leftrightarrow \mathfrak{u}(1)_Y$.

Sub-tasks (i) and (iii) produce Phase Misalignments (PM-ZC22-S01, PM-ZC22-III-01). LEM-ZC22-01: $C_{jk}^l = 0$ identically (additive bracket fails for abelian group).

Architectural discovery (FIND-ZC22-S01): $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ structure constants require tensor product interaction $(\mathbb{Z}_3)^\times \otimes (\mathbb{Z}_5)^\times$. CONJ-ZC21-01 remains Candidate pending ZC22-ext.

5.4 ZC23 — G-Partition and the Non-Abelian χ -Map (Part XXIV)

ZC23: DEF-ZC23-01 and CONJ-ZC23-01

Following the tensor product discovery of ZC22, this paper formalises the G-Partition Non-Abelian Chi-Map χ_{NA} (DEF-ZC23-01), which assigns each δ -event to the $\text{SU}(2)$ channel with probability $\mathcal{G}_{\text{coup}} = \ln 2/D_0 \approx 0.7374$.

Observable $f_{\text{RMS}}(N)$ converges to plateau $\sqrt{3} \approx 1.732$ as $N \rightarrow \infty$ with power-law exponent $\alpha = 0.001 \pm 0.004$ (simulator V22.3, T-canonical gauge).

Six Phase Misalignments registered (PM-ZC23-01.06) as scar tissue. CONJ-ZC23-01 placed on firm candidate status. FIND-ZC23-01: $1 + w_{\text{DE}} = \mathcal{G}/d_s \approx 0.246$.

5.5 ZC24 — Fine Structure Constant from the δ -Chain (Part XXIV)

ZC24: $\alpha_{\text{EM}}^{R_0} = (8/15)\Gamma_{\text{CRF}}$

TASK-ZC23 established $1 + w_{\text{DE}} = \mathcal{G}/d_s$ and TASK-ZC20 established $\cos^2 \theta_W = \ln n / \ln |Z_{15}|$. Their sum $\cos^2 \theta_W + (1 + w_{\text{DE}}) = 1.01367$ falls 1.37% short of exact complementarity.

Main discovery: Defining $\Gamma_{\text{CRF}} := \mathcal{G}/d_s - \sin^2 \theta_W = 0.013664$, the relation $\alpha_{\text{EM}} \approx (\varphi(15)/15) \Gamma_{\text{CRF}} = (8/15) \Gamma_{\text{CRF}}$ holds at gap +0.132%.

Complementarity gap 1.37% decomposes into three layers: algebraic shortfall $\sim 1.29\%$ (algebraic independence of $\ln 2, \ln 3, \ln 5$); D_0 convention term 0.048%; residual 0.035%. Gap 1.37% is the cost of integer quantization, not an error.

5.6 ZC28-v3 — Chi-Map Coupling Rate Γ_{CRF} (Part XXIV)

ZC28-v3: THM-ZC28-01 — Near-Formal Closure

This is the registry-completed revision of TASK-ZC28 v2. v3 fixes: COR-ZC28-01 given explicit corollary environment; ratio values in FIND-ZC28-03 corrected; Summary item 5 clarified.

THM-ZC28-01: $\Gamma_{\text{CRF}} = \mathcal{G}/d_s - \sin^2 \theta_W$ (near-formal closure of the full parameter-free α_{EM} chain). GAP-ZC28-01 opened: 3.49% ε_0 reconciliation gap, target for ZC29+.

5.7 ZC25-v2 — Propagator Weight and Force Hierarchy Conjecture (Part XXV)

ZC25-v2: R_0 -layer Identification and GAP-ZC25-03

TASK-ZC24 established $\alpha_{\text{EM}} \approx (8/15)\Gamma_{\text{CRF}}$ at gap +0.132%.

Direction B: Argument for 8/15 decomposes into B1 (uniform distribution, proved by THM-TC01) and B2 (EM-active = $\mathbb{G}_{\text{gen}}$, near-formal from coprimality). Force hierarchy $\alpha_i = (|S_i|/15)\Gamma_{\text{CRF}}$ as conditional prediction (CONJ-ZC25-B02).

R-layer analysis (v2): All Z-Programme quantities — D_0 , $\mathcal{G}$, $\cos^2 \theta_W$, w_{DE} , Γ_{CRF} — are R_0 -layer (receiver-independent, pre-resolution) quantities. Gap 0.132% decomposes into: θ -nonlinearity term (-2.65×10^{-5}) and R_1 -wave shift ($+3.42 \times 10^{-5}$). The R_1 -wave shift = $0.600\varepsilon_0^2$ is the precise target for ZC26.

5.8 ZC26 — R_1 Wave Crossing: Back-Calculation of ε_0 (Part XXV)

ZC26: DEF-ZC26-RS01 and GAP-ZC26-01

TASK-ZC25 v2 showed gap 0.132% is a layer-crossing cost and opened GAP-ZC25-03 (the R_1 -wave shift $0.600\varepsilon_0^2$).

Part A: Systematic probe finds 0.600 has no clean form in current CRF. GAP-ZC26-01 registers the need for a formal theory of EM photon propagation through $\mathbb{G}_{\text{gen}}$ channels in the R_1 layer.

Part B (Resolution-Scale Protocol): DEF-ZC26-RS01 formalises a five-item checklist that must be applied before any quantity is claimed to “match” or “gap from” experiment. This is a governance-level output addressing PM-ZC26-01 and predecessors.

5.9 ZC27 — R_1 Channel Dynamics via φ -Cancellation (Part XXV)

ZC27: THM-ZC27-01 — CONJ-ZC26-01 Promoted

TASK-ZC26 established the target $\delta_{\text{DKL}}^{R_1} = 3.421 \times 10^{-5} = 0.600\varepsilon_0^2 = (d_s/n_m)\varepsilon_0^2$ (CONJ-ZC26-01) and registered GAP-ZC26-01.

This paper closes GAP-ZC26-01. Using DEF-RN02 ($T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$) and CRT applied to $\mathbb{G}_{\text{gen}} = (\mathbb{Z}/15\mathbb{Z})^*$: $\Sigma_{\text{inv}}^{\text{EM}} = d_s/\varphi(15)$ and $dt^{\text{EM}} = \varphi(15)/n_m$. Product: $\Sigma_{\text{inv}}^{\text{EM}} \cdot dt^{\text{EM}} = d_s/n_m$ exactly ($\varphi(15)$ cancels).

THM-ZC27-01: $\delta_{\text{DKL}}^{R_1} = (d_s/n_m)\varepsilon_0^2$ (numerical gap 0.025%). CONJ-ZC26-01 promoted. Only the prime factorisation $15 = d_s \cdot n_m$ matters.

5.10 ZC29 — Force Hierarchy Theorem (Part XXVI)

ZC29: THM-ZC29-01 — Universal Coupling Rule

TASK-ZC24 established $\alpha_{\text{EM}}^{R_0} = (8/15)\Gamma_{\text{CRF}}$ and CONJ-ZC25-B02 conjectured the rule extends to all three forces. This paper proves that conjecture.

$\mathbb{Z}_{15}$ generator partition: $\mathbb{G}_{\text{gen}}$ ($|8|$, EM), G_5 ($|4|$, Weak), G_3 ($|2|$, Strong), $\{0\}$ (Vacuum).

THM-ZC29-01: For sector $S_i \subset \mathbb{Z}_{15}$ of order d_i : $\alpha_i^{R_0} = (\varphi(d_i)/15)\Gamma_{\text{CRF}}$. Exact ratios: $\alpha_{\text{Weak}}/\alpha_{\text{EM}} = 1/2$; $\alpha_{\text{Strong}}/\alpha_{\text{EM}} = 1/4$.

FIND-ZC29-03: For confined sectors ($d_i < 15$), net rate $\Gamma_i < 0$ — the CRF signature of confinement. Open gaps: GAP-ZC29-01 ($R_0 \rightarrow R_{\text{max}}$ running coupling crossing cost); GAP-ZC29-02 (gravity sector from vacuum element).

5.11 ZC30 — Sector Crossing Costs: Universal φ -Cancellation (Part XXVI)

ZC30: THM-ZC30-01/02 — GAP-ZC29-01 Closed

TASK-ZC27 derived $T(R_1)_{\text{EM}} = (d_s/n_m)\varepsilon_0^2$ by $\varphi(15)$ -cancellation. This paper closes GAP-ZC29-01 for Weak and Strong sectors.

THM-ZC30-01: Weak sector ($G_5 \subset \mathbb{Z}_5$): $T(R_1)_{\text{Weak}} = (n_m/d_s)\varepsilon_0^2$.

THM-ZC30-02: Strong sector ($G_3 \subset \mathbb{Z}_3$): $T(R_1)_{\text{Strong}} = (d_s/n_m)\varepsilon_0^2$.

FIND-ZC30-01 (φ -universality): $\varphi(d_i)$ cancels in $T(R_1)_i$ for *every* sector. Crossing cost depends only on the ratio of generator modes to receiver modes.

FIND-ZC30-02: Two classes — Spatial-type (EM, Strong): $T(R_1) = (d_s/n_m)\varepsilon_0^2$ (massless bosons); Matter-type (Weak): $T(R_1) = (n_m/d_s)\varepsilon_0^2$ (massive bosons). Ratio: $T(R_1)_{\text{Weak}}/T(R_1)_{\text{EM}} = (n_m/d_s)^2 = 25/9$ (exact).

Quick-Access Reference: Open Gaps as of v17.6

Gap ID	Description	Status/Owner
GAP-MC-01	Residual 10–17% in lepton mass ratios	Open (ZC16)
GAP-ZC17-01	Absolute Yukawa scale	Conditionally closed by ZC17 given v input
GAP-ZC17-02	EW gauge group $SU(2)_L \times U(1)_Y$ from δ -chain	Closed at mixing-angle level (ZC20)
GAP-ZC17-03	Hierarchy $v/m_{\text{Pl}} \sim 10^{-17}$	Near-closed (ZC19: +0.01%; ZC20 confirmed)
GAP-ZC21-00	Continuous Lie group from discrete $\mathbb{Z}_{15}$ in $N \rightarrow \infty$	Open; ZC22-ext planned
GAP-ZC23-TH	Formal proof THM-ZC23-01 for $f_{\text{RMS}} \rightarrow \sqrt{3}$	Closed (Part E ZC33 v2 CLT proof)
GAP-ZC25-02	Γ_{CRF} formal construction	Near-closed (ZC28-v3 THM-ZC28-01)
GAP-ZC26-01	Formal R_1 -wave theory for EM sector	Closed by ZC27 (THM-ZC27-01)
GAP-ZC28-01	3.49% ε_0 reconciliation gap	Closed (Part E ZC37 THM-ZC37-01)
GAP-ZC29-01	$R_0 \rightarrow R_{\text{max}}$ running coupling crossing cost	Closed for EM/Weak/Strong (ZC30)
GAP-ZC29-02	Gravity sector identification from vacuum element	Open
GAP-ZC29-03 / ZC30-02	Strong-sector witness (“adjacent”→“orthogonal” correction)	PM-ZC29-04 registered
GAP-ZC24-COMP-01	Complementarity gap +1.37%	Closed (Part E ZC34 THM-ZC34-01)
GAP-ZC29-03	$R_0 \rightarrow R_1$ semigroup dynamics formal	Closed (Part E ZC33 THM-ZC33-01/02)
GAP-ZC30-01	Sector-specific $\varepsilon_0^{(i)}$	Closed (Part E ZC37 COR-ZC37-03)
GAP-ZC35-01	Number-theoretic proof: $\mathbb{Z}_{15}$ unique for binary-ladder totients	Open (Part E ZC35)
GAP-ZC36-01	Exact derivation of K14 ($F\varphi_{\text{gold}}^2 = 4$)	Near-closed (Part E ZC39 gap −0.0034%; ZC40 THM-ZC40-01)
GAP-ZC36-02	Structural gap between Route A and Route C (0.21%)	Open (Part E ZC36)
GAP-ZC38-02	K14 exact consistency with K7'	Near-closed (Part E ZC39/40: under exact K7', 0.000%)
GAP-ZC39-01	Why $1/\varphi_{\text{gold}}$ bleeding ratio at R_1 crossing	Closed (Part E ZC40 LEM-ZC40-01)

6 ZC31–ZC40 Abstract Registry (Standalone Papers, Not Yet Integrated)

Integration Status

The ten papers below (ZC31–ZC40) are standalone files in the project. They are *not yet* integrated into Parts A–D of CRF Complete. Their abstracts are reproduced here to enable Pre-Work Audit without opening each file. Integration will occur at the next Complete Edition update cycle (v17.7 planned).

Key closures in this cluster: GAP-ZC24-COMP-01 (ZC34), GAP-ZC28-01 (ZC37), GAP-ZC29-03 (ZC33), GAP-ZC30-01 (ZC37), GAP-ZC39-01 (ZC40). K14 elevated to Near-Formal Candidate (ZC39/ZC40). I_{eff} first structural derivation from δ -chain (ZC38).

6.1 ZC31 — Confinement Spectrum Closed Form

ZC31: FIND-ZC31-01/02/03 — Algebraic Spectrum + Weinberg Bridge

TASK-ZC29 established the sector net rate Γ_i and FIND-ZC29-03 correlated its sign with confinement. This paper closes the algebraic scope: a post-ZC30 Council session identified the destruction rate as a Shannon information-loss fraction

$$R_{\text{destroy},i} = 1 - \frac{\ln \varphi(d_i)}{\ln 15}, \quad \text{DEF-ZC29-04.}$$

Three exact closed forms follow:

$$\text{FIND-ZC31-01: } \Gamma_i = \mathcal{G}_{\text{coup}}/d_s - 1 + \ln \varphi(d_i)/\ln 15,$$

$$\text{FIND-ZC31-02: } \Gamma_{\text{vac}} = \mathcal{G}_{\text{coup}}/d_s - 1 \quad (\text{spectrum floor, exact}),$$

$$\text{FIND-ZC31-03: } \Gamma_{\text{EM}} = \mathcal{G}_{\text{coup}}/d_s - 1 + \sin^2 \theta_W.$$

FIND-ZC31-03 is a Weinberg bridge: $\sin^2 \theta_W = \ln 8 / \ln 15$ (ZC20) reappears as the EM-sector recovery from vacuum. PM-ZC34-01 ($\cos^2 \theta_W$ / $\sin^2 \theta_W$ convention) noted. *Convention Bridge Note (CRF-GP-2026-002): the displayed FIND-ZC31-03 equation above retains its pre-PM-ZC34 legacy form, in which the quantity $\ln 8 / \ln 15 \approx 0.7679$ carries the label $\sin^2 \theta_W$. Under the PM-ZC34-01 locked convention this quantity is $\cos^2 \theta_W$ (LARGE), and $\sin^2 \theta_W \approx 0.23213$ (SMALL). The legacy “+ $\sin^2 \theta_W$ ” must be read as “+ $\cos^2 \theta_W$ ” to reproduce $\Gamma_{\text{EM}} = +0.013664$. Source equation preserved per the additive/Failure-Stance rule.*

6.2 ZC32 — Operator Form of the Confinement Spectrum

ZC32-v2: THM-ZC32-01 — M_i Spectral Decomposition; PM-ZC32-01

TASK-ZC31 held CONJ-ZC29-05 (eigenvalue interpretation of Γ_i) for this paper.

Result 1 (falsification). CONJ-ZC29-05 as literally stated is *falsified*: neither the ZC20 preimage form nor the ZC23 stochastic form of χ carries a dynamical propagator with eigenvalue spectrum. Registered as **PM-ZC32-01** (scar tissue).

Result 2 (surviving structure). Define

$$M_i := R_{\text{create}} \cdot I - R_{\text{destroy},i} \cdot \pi_i,$$

where $\pi_i : L^2(\mathbb{Z}_{15}) \rightarrow L^2(\mathbb{Z}_{d_i})$ is the orthogonal projection (**DEF-ZC32-01**). **THM-ZC32-01**: Γ_i is an eigenvalue of M_i on the sector subspace. The propagator structure is real and sector-local. **CONJ-ZC32-01**: the $R_0 \rightarrow R_1$ lifting of M_i as a semigroup generator (target for ZC33).

6.3 ZC33 — R_1 -Semigroup Dynamics from Existing Pillars

ZC33-v2: THM-ZC33-01/02 — GAP-ZC29-03 Closed; Sector Physical Time

TASK-ZC32 sharpened GAP-ZC29-03 into a four-item roadmap. This paper closes all four items using only existing CRF objects.

DEF-ZC33-01: Sector physical time $t_i := (p_i/q_i) \cdot T_{\text{avg}}$, where $p_i/q_i = \Sigma_{\text{inv}}^{(i)} \cdot dt^{(i)}$ from ZC27/ZC30, and $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$ from Part B (Wald + Jaynes). No new constant introduced.

THM-ZC33-01: The R_1 -wave generator on sector subspace is $A_i^{R_1} = T_{\text{avg}}^{-1} M_i$ (restricted). The $R_0 \rightarrow R_1$ crossing of CONJ-ZC32-01 is explicit and grounded.

THM-ZC33-02: Correlation decay $\langle f(t), f(0) \rangle = e^{\Gamma_i t} \|f(0)\|^2$ on sector subspace. Sector physical time t_i sets the semigroup clock. GAP-ZC29-03 **closed**.

Pillars used: DEF-RN02 ($T(R_1) = \varepsilon_0^2 \Sigma_{\text{inv}} dt$), FIND-ZC30-01 (φ -universality), Part B $T_{\text{avg}} + \text{Jaynes}$, $\lambda_2 T_{\text{avg}} = n - 1$.

6.4 ZC34 — Complementarity Identity: $\cos^2 \theta_W + (1 + w_{\text{DE}}) = 1 + \Gamma_{\text{EM}}$

ZC34: THM-ZC34-01 — GAP-ZC24-COMP-01 Closed; PM-ZC34-01

TASK-ZC24 registered GAP-ZC24-COMP-01: the complementarity residual $\cos^2 \theta_W + (1 + w_{\text{DE}}) = 1.01367$ (+1.37%) declared “cannot be closed within the current δ -chain.” A post-ZC33 corpus audit identified the residual = Γ_{EM} (FIND-ZC31-03). Three-line algebraic closure:

$$\cos^2 \theta_W + (1 + w_{\text{DE}}) = 1 + \Gamma_{\text{EM}} \quad (\text{THM-ZC34-01, exact}).$$

The “gap” is the CRF-predicted cosmological order-creation rate — not an error. Three additional identities from the same $\mathbb{Z}_{15}$ partition:

$$\text{THM-ZC34-02: } \sum_i \alpha_i^{R_0} = \Gamma_{\text{CRF}} \quad (\text{Gauss divisor sum: } \sum_{d|15} \varphi(d) = 15)$$

$$\text{THM-ZC34-03: } \alpha_{\text{EM}}^{R_0} + \Gamma_{\text{vac}} = \mathcal{G}_{\text{coup}}/d_s \quad (\text{creation-rate conservation})$$

$$\text{THM-ZC34-04: } \cos^2 \theta_W - \sin^2 \theta_W = \Gamma_{\text{EM}} - \Gamma_{\text{vac}} \quad (\text{mixing-angle-confinement duality})$$

PM-ZC34-01: Weinberg-angle convention lock ($\cos^2 \theta_W = \ln 8 / \ln 15 \approx 0.7679$ LARGE; $\sin^2 \theta_W \approx 0.2321$ SMALL). GAP-ZC24-COMP-01 **closed**.

6.5 ZC35 — Spectral Sum Identities

ZC35: THM-ZC35-01 — Spectral Sum; OBS-ZC35-01; GAP-ZC35-01

Combining FIND-ZC31-01, FIND-ZC23-01, the totient log-sum, and THM-ZC20-01 (PM-ZC34-01 convention) in one algebraic step:

$$\sum_i \Gamma_i = 4 \left(\frac{\mathcal{G}_{\text{coup}}}{d_s} - 1 \right) + 2 \cos^2 \theta_W \quad (\text{THM-ZC35-01, exact}).$$

FIND-ZC35-01: Recovery lattice recovery_i = $k_i \cdot \cos^2 \theta_W / 3$ for $k_i \in \{0, 1, 2, 3\}$ — arithmetic progression with step $\ln 2 / \ln 15$.

FIND-ZC35-02: Totient log-sum identity $\sum_{d|15} \ln \varphi(d) = 2 \ln n_{\text{CRF}}$, where $n_{\text{CRF}} = 2^{d_s} = 8$ is the CRF Key Identity constant.

OBS-ZC35-01: $\mathbb{Z}_{15}$ is the unique $n \leq 100$ with $\tau(n) = 4$ whose totient values form a strict binary ladder $\{1, 2, 4, 8\}$.

GAP-ZC35-01: Number-theoretic proof valid for all n (open).

6.6 ZC36 — Fiedler Value and Golden Ratio: Three Routes

ZC36: THM-ZC36-01, CONJ-ZC36-01 ($F\varphi_{\text{gold}}^2 = 4$), COR-ZC36-01

The Fiedler value F of the R_1 -channel KL-graph of $\mathbb{Z}_{15}$ ($F_{\text{meas}} = 1.527$ in TLS v3) admits three algebraic routes:

Route A (Jaynes–Wald composition, THM-ZC36-01): $F = 2nK^{*2}/[e(n-1)\varepsilon_0] = 1.524587$ (gap -0.158%). Rearrangement of Jaynes ($K^*T_{\text{avg}}/F = e$, Part B K20) and Wald ($T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$, Part B K35).

Route B (WaveChain TLS v3): $F = (3/2 - \sqrt{d_s}/n^2)(1 + \sigma_\Psi / \ln \varphi_{\text{gold}}) = 1.52583$ (gap -0.077%). Part B citation; not lifted as ZC theorem.

Route C (Fibonacci limit, CONJ-ZC36-01): $F = 4/\varphi_{\text{gold}}^2$; numerical 1.527864 (gap $+0.057\%$). This is Part B K14 hypothesis $F\varphi_{\text{gold}}^2 = 4$, lifted to ZC.

COR-ZC36-01: $\Gamma_{\text{EM}} \equiv \Gamma_{\text{CRF}}$ (one-line consequence of FIND-ZC31-01 + DEF-ZC24-GAMMA-01; first explicit registration).

FIND-ZC36-01: $\prod_{d|15} \varphi(d) = 64 = 2^{T(d_s)}$ (T = triangular number).

FIND-ZC36-02: $\prod_{d|15} d = 225 = |\mathbb{Z}_{15}|^2$.

GAP-ZC36-02: 0.21% structural gap between Route A and Route C (open).

PM-ZC36-01: Pre-Work Audit false alarm — Council scan pointed to “new scar in ZC31” but PM-ZC34-01 already covered it in the parent.

6.7 ZC37 — Epsilon Floor: Closing GAP-ZC28-01

ZC37-v1.2: THM-ZC37-01 — $\varepsilon_{\text{EM}} = \varepsilon_{\text{can}}$; Three Gap Closures

Three gaps (GAP-ZC27-01, GAP-ZC28-01, GAP-ZC30-01) traced to a single registration error: DEF-ZC28-02 had identified $\varepsilon_{\text{EM}} := \alpha_{\text{EM}}^{R_0}$, but these are categorically distinct. $\alpha_{\text{EM}}^{R_0} = (8/15)\Gamma_{\text{CRF}}$ carries a $\mathbb{Z}_{15}$ sector weight; $\varepsilon_{\text{can}} = |I_{\text{eff}} - \ln 2| = 0.007553$ is the canonical TLS instability floor, pre-sector, universal.

THM-ZC37-01: $\varepsilon_{\text{EM}} = \varepsilon_{\text{can}} = |I_{\text{eff}} - \ln 2|$. Under this identification $T(R_1)_{\text{EM}} = (d_s/n_m)\varepsilon_{\text{can}}^2 = 3.4227 \times 10^{-5}$ (ZC26 target: 3.421×10^{-5} ; gap $+0.050\%$).

COR-ZC37-01: GAP-ZC28-01 (3.49% discrepancy) **closed**.
COR-ZC37-02: THM-ZC27-01 is unconditional under THM-ZC37-01.
COR-ZC37-03: GAP-ZC30-01 (sector-specific $\varepsilon_0^{(i)}$) **closed** trivially via FIND-ZC30-01.
FIND-ZC37-01: Gift II-1 diagnostic ($\varepsilon_{\text{EM}} = \alpha_{R_0} \cdot (15/8)^{1/16}$, gap 0.38%) confirmed that α_{R_0} and ε_0 are different objects.
FIND-ZC37-02 (v1.2 post-pub): Forward connection to ZC38 (I_{eff} derivation from δ -chain).

6.8 ZC38 — First Structural Derivation of I_{eff}

ZC38: CONJ-ZC38-01 — I_{eff} from δ -Chain; Gap +0.0066%

Since TASK-ZC18 Track A, $I_{\text{eff}} = 0.7007$ has been a fixed empirical constant of the TLS. TASK-ZC37 explicitly registered its derivation as “the fundamental open problem of the TLS architecture.”

The key observation: the Jaynes identity ($K^*T_{\text{avg}}/F = e$, Part B K20) and K14 ($F\varphi_{\text{gold}}^2 = 4$, Part B) form a self-consistency equation. Substituting Wald’s T_{avg} into Jaynes and eliminating F via K14:

$$\varepsilon_0 = \frac{n(1 - e^{-1/n})^2 \varphi_{\text{gold}}^2}{2(n-1)e}, \quad I_{\text{eff}} = \ln 2 + \varepsilon_0. \quad (\text{CONJ-ZC38-01})$$

All quantities determined by δ -chain alone: $n = 8$, φ_{gold} , e .

Numerical: $I_{\text{eff}}^{\text{ZC38}} = 0.700746$; gap +0.0066% from SINDy value — among the tightest derivations in the Z-Programme. Gap in ε_0 : +0.61% (combined Jaynes 0.119% + K14 0.133% residuals).

S_{dep} interpretation confirmed: $\varepsilon_0 > 0$ follows necessarily from $n > 1$ and $\varphi_{\text{gold}} > 0$. GAP-ZC38-02 (K14 exact consistency) inherited from ZC36.

6.9 ZC39 — K14 Renormalization: Near-Formal Closure

ZC39: CONJ-ZC39-01 — $F_{\text{ren}}\varphi_{\text{gold}}^2 = 3.99986$; K14 Near-Formal

K14 ($F\varphi_{\text{gold}}^2 = 4$) has been a confirmed Hypothesis since Part B (gap -0.013% to -0.133%). Part B showed K14 cannot be derived exactly from the current WaveChain formulas (K7' + K8): this was GAP-ZC36-01 / GAP-ZC38-02.

Missing ingredient: K8 ($\sigma_K = \varepsilon_0(n+2)/\sqrt{2n+3}$) gives the R_0 -layer AR(1) variance, omitting the R_1 -crossing contribution from THM-ZC27-01 ($T_{R_1} = (d_s/n_m)\varepsilon_0^2$).

Renormalized variance (**CONJ-ZC39-01**):

$$(\sigma_R)^2 = (\sigma_K)^2 + \frac{1}{\varphi_{\text{gold}}} T_{R_1} = \varepsilon_0^2 \left[\frac{(n+2)^2}{2n+3} + \frac{d_s}{\varphi_{\text{gold}} n_m} \right].$$

The golden fraction $1/\varphi_{\text{gold}} = \varphi_{\text{gold}} - 1$ is the Fibonacci bleeding ratio.

Result: $F_{\text{ren}}\varphi_{\text{gold}}^2 = 3.99986$ (gap -0.0034%) — $36\times$ improvement over K8 alone. K14 elevated to **Near-Formal Candidate**.

FIND-ZC39-01: ZC38 + ZC39 mutually consistent; Jaynes + F_{ren} pins $\varepsilon_0 = 0.007599$ (matches CONJ-ZC38-01).

6.10 ZC40 — Fibonacci Partition: Why $1/\varphi_{\text{gold}}$ at R_1 Crossing

ZC40: THM-ZC40-01 — CONJ-ZC39-01 Promoted; GAP-ZC39-01 Closed

TASK-ZC39 left GAP-ZC39-01 open: *why* exactly $1/\varphi_{\text{gold}}$ of T_{R_1} contributes to σ_{Ψ}^2 .

LEM-ZC40-01 (Fibonacci Energy Conservation): At any R_1 crossing, wave energy T_{R_1} partitions into exactly two channels in the golden ratio:

$$T_{R_1} = \underbrace{\frac{1}{\varphi_{\text{gold}}} T_{R_1}}_{E_{\text{var}}} + \underbrace{\frac{1}{\varphi_{\text{gold}}^2} T_{R_1}}_{E_{\text{amp}}}, \quad 1/\varphi_{\text{gold}} + 1/\varphi_{\text{gold}}^2 = 1 \text{ (Fibonacci identity)}.$$

$E_{\text{var}} \rightarrow \Psi$ field variance (entropic channel); $E_{\text{amp}} \rightarrow$ WaveChain amplitude F (coherent channel). The φ_{gold} -split is the *unique* partition consistent with energy conservation and the φ_{gold} -architecture ($F\varphi_{\text{gold}}^2 = 4$, K14).

LEM-ZC40-02 (AR(1) Shift): $\sigma_{\Psi}^2(R_1) = \sigma_{\Psi}^2(R_0) + E_{\text{var}}$.

THM-ZC40-01: CONJ-ZC39-01 promoted to theorem. $F\varphi_{\text{gold}}^2 = 3.99986$ (gap -0.0034%). GAP-ZC39-01 **closed**.

FIND-ZC40-01: Exact fraction $f_{\text{exact}} = 0.636$ giving $F\varphi_{\text{gold}}^2 = 4.000$ exactly; differs from $1/\varphi_{\text{gold}} = 0.618$ by 2.9% , consistent with K7' approximation gap. Under exact K7', K14 closes to 0.000% .

ZC Papers Now Integrated into Part E (v17.7)

The following standalone ZC papers, registered above as “Not Yet Integrated” in v17.6, are now integrated into **Part E (v17.7)** verbatim. Their standalone source files in the project remain unchanged (CUIP No-Loss compliance).

- CRF_TASK_ZC31_ConfinementSpectrum_v1.tex — Integrated in Part E, Part XXVII, Section ZC31
- CRF_TASK_ZC32_OperatorForm_v2.tex — Integrated in Part E, Part XXVII, Section ZC32
- CRF_TASK_ZC33_R1Semigroup_v2.tex — Integrated in Part E, Part XXVII, Section ZC33
- CRF_TASK_ZC34_ComplementarityIdentity_v1.tex — Integrated in Part E, Part XXVIII, Section ZC34
- CRF_TASK_ZC35_SpectralSumIdentities_v1.tex — Integrated in Part E, Part XXVIII, Section ZC35
- CRF_TASK_ZC36_FiedlerGoldenRatio_v1.tex — Integrated in Part E, Part XXIX, Section ZC36
- CRF_TASK_ZC37_EpsilonFloor_v1_2.tex — Integrated in Part E, Part XXIX, Section ZC37
- CRF_TASK_ZC38_IeffFloor_v3.tex — Integrated in Part E, Part XXIX, Section ZC38
- CRF_TASK_ZC39_K14Renorm_v1.tex — Integrated in Part E, Part XXIX, Section ZC39
- CRF_TASK_ZC40_FibonacciPartition_v1.tex — Integrated in Part E, Part XXIX, Section ZC40

Cross-Part Connection Map for these ten papers is given in Part E, **Part XXX** — a NEW section type introduced in v17.7.

7 Part E — Cluster Closures & Cross-Part Map (Parts XXVII–XXX)

Volume Abstract

Part E — CRF Complete Edition v17.7 (Parts XXVII–XXX)

This is **Part E** of the CRF Complete Edition v17.7, the fifth of five sequential files comprising one continuous volume on the δ -mechanism. Part E integrates the ten post-v17.6 standalone papers (ZC31–ZC40, May 2026) verbatim, plus introduces a new section type — the Cross-Part Connection Map (Part XXX).

Structural ascent of Part E:

- **Part XXVII** (Confinement Operator and R_1 Semigroup): ZC31 promotes the Council information-loss form to atomic units; ZC32 lifts Γ_i to an operator M_i on $L^2(\mathbb{Z}_{15})$; ZC33 bridges to the R_1 -semigroup via $A_i^{R_1} = T_{\text{avg}}^{-1}M_i$. Closes GAP-ZC29-03.
- **Part XXVIII** (Complementarity and Spectral Sum): ZC34 closes GAP-ZC24-COMP-01 (reframing: residual = Γ_{EM}). ZC35 gives the closed spectral sum $\sum \Gamma_i = 4(\mathcal{G}_{\text{coup}}/d_s - 1) + 2\cos^2\theta_W$ with the recovery lattice and totient log-sum identity.
- **Part XXIX** (TLS Layer Closures): ZC36 (Fiedler, three routes); ZC37 ($\varepsilon_{\text{EM}} = \varepsilon_{\text{can}}$, closes GAP-ZC28-01 and GAP-ZC30-01); ZC38 (first structural I_{eff} , gap 0.0066%); ZC39 (K14 renormalization); ZC40 (Fibonacci partition, GAP-ZC39-01 closed, K14 promoted to Near-Formal Theorem).
- **Part XXX** (Cross-Part Connection Map): NEW SECTION TYPE in v17.7. For each ZC31..40 paper, records backward inheritance from Parts A/B/C/D, forward correction of Parts A/B/C/D interpretations (no source-file edit), and cross-Part Phase Misalignments. Includes a Master Z-Programme Status Table v17.7 delta and a Pre-Work Audit Protocol summary (PWA-1..PWA-4).

Key closures in Part E: GAP-ZC24-COMP-01 (ZC34), GAP-ZC28-01 (ZC37), GAP-ZC29-03 (ZC33), GAP-ZC30-01 (ZC37), GAP-ZC39-01 (ZC40), GAP-ZC36-01/GAP-ZC38-02 (Near-closed by ZC39/40). K14 hypothesis elevated to Near-Formal Theorem. I_{eff} first structural derivation.

Phase Misalignments preserved: PM-ZC32-01, PM-ZC32-02, PM-ZC33-01, PM-ZC34-01, PM-ZC36-01, PM-ZC37-01.

Pre-Work Audit lessons codified: PWA-1 (R-layer scan before R_0/R_1 writing), PWA-2 (combine existing pillars before inventing), PWA-3 (Complete Edition Index scan for foundational machinery), PWA-4 (parent PM-registry scan before flagging child scars).

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8 Part F — ZC41–44 Cluster, Mind/Barami, Axiom 11 (Parts XXXI–XXXIV)

Volume Abstract

Part F — CRF Complete Edition v17.8 (Parts XXXI–XXXIV)

This is **Part F** of the CRF Complete Edition v17.8, the sixth of six sequential files. Part F integrates nine post-v17.7 standalone papers in four new Parts.

Structural ascent of Part F:

- **Part XXXI** (ZC41–44 Cluster): ZC41 closes GAP-ZC40-01 and promotes K14 to exact unconditional Theorem (THM-ZC41-01; $F\varphi_{\text{gold}}^2 = 4$ exact). ZC42 derives the Born-chain spectral gap λ_2 via Dirichlet + n_m -sector suppression (THM-ZC42-01, gap 0.77%). ZC43 proves $n = 15$ is the unique integer with $\tau(n) = 4$ and binary-ladder totients (THM-ZC43-01; closes GAP-ZC35-01). ZC44 derives the bridge identity $\mathcal{R} = d_s n_m / n = 15/8$ in four exact forms (FIND-ZC44-01), closes GAP-ZC42-01 via CRT chain (THM-ZC44-01), and registers the three-axis $\mathbb{Z}_{15}$ picture (CONJ-ZC44-01).
- **Part XXXII** (Mind/Barami Layer): Mind/Barami Companion v2 extends the Agentive-mode structure. Bridge Extension Φ formalises DEF-UF- Φ 01. Gifts HIJ registers CONJ-GH-01, promotes CONJ-GI-01, and registers FIND-GJ-01/GAP-GJ-01. Gift K completes the B_+ -observable framework.
- **Part XXXIII** (Axiom 11 Upgrade): PM-AX11-01 preserved as scar tissue. Axiom 11 v1 retired (RETIRED-AS-STATED). Five-component upgrade following PM-ZC21-01 pattern.
- **Part XXXIV** (Cross-Part Connection Map v17.8): Extends Part XXX with records for all nine Part F papers. Adds PWA-5 (mechanism-count audit protocol).

Key closures: GAP-ZC35-01 (ZC43), GAP-ZC40-01 (ZC41), GAP-ZC41-01 (ZC41 self), GAP-ZC42-01 (ZC44), GAP-MB02 and GAP-UF- Φ 01 (conditional, CONJ-GI-01).

Phase Misalignments preserved: PM-AX11-01 (Axiom 11 upgrade), PM-ZC44-01 (M5/M7 CRT infrastructure overlap), OBS-ZC44-01 (mechanism count = $\varphi(15) = 8$, open bijection).

Pre-Work Audit lessons: PWA-5 (mechanism-count audit: enumerate, check pairwise

shared infrastructure, declare independence at conclusion level, register overlaps before promoting).

Contents Map

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9 Part G — ZC45–49 Cluster, $\mathbb{Z}_{15}$ Projection Chain, Four-Axis Duality (Parts XXXV–XXXVII)

Volume Abstract

Part G — CRF Complete Edition v17.9 (Parts XXXV–XXXVII)

This is **Part G** of the CRF Complete Edition v17.9, the seventh of seven sequential files. Part G integrates five post-v17.8 standalone papers in three new Parts.

Structural ascent of Part G:

- **Part XXXV** (ZC45 — Universal Floor): Closes GAP-GJ-01 by proving THM-ZC45-01: the instability floor ε_0 is identical in both CRF operating modes — $\varepsilon_{0,\text{Spont}} = \varepsilon_{0,\text{Agent}} = \varepsilon_0$. COR-ZC45-01 promotes FIND-GJ-01 to Theorem. PM-ZC45-01 preserves the 0.16% Wald residual (GAP-ZC42-02, inherited). First paper to formally unify the ε_0 floor across both Physics and Mind strata.
- **Part XXXVI** (ZC46–49 Cluster — $\mathbb{Z}_{15}$ Projection Chain): ZC46 promotes CONJ-ZC44-01 to THM-ZC46-01 (Three-Projection Theorem): $\mathcal{R} = 15/8$ is the common algebraic root of three $\mathbb{Z}_{15}$ projections (ZC18-B totient axis, ZC42 CRT axis, ZC43 divisor axis). GAP-ZC46-01 registered. ZC47 closes GAP-ZC46-01 via THM-ZC47-01 (CRT Sector Decoupling): tensor decomposition $\ell^2(\mathbb{Z}_{15}) \cong \ell^2(\mathbb{Z}_3) \otimes \ell^2(\mathbb{Z}_5)$; THM-ZC46-01 becomes unconditional (COR-ZC47-02). CONJ-ZC47-01 registered. ZC48 proves CONJ-ZC47-01 exactly via CRF Model A ($\tau_i \propto 1/\alpha_i^{R_0}$, LEM-ZC48-01) tracing to THM-ZC29-01; THM-ZC48-01: $\tau_{\mathbb{Z}_3}/\tau_{\mathbb{Z}_5} = \varphi(5)/\varphi(3) = 2$ exact. ZC49 synthesises four axes of $\mathbb{Z}_{15}$ sector structure into THM-ZC49-01 (Four-Axis Duality); FIND-ZC49-04 reveals the Key Identity in totient disguise.
- **Part XXXVII** (Cross-Part Connection Map v17.9): Extends Part XXXIV with backward inheritance, forward corrections, and Phase Misalignment records for ZC45–ZC49. Adds PWA-6 (macro convention cross-check before embedding any new ZC paper). Includes Master Z-Programme Status Table v17.9 delta.

Key closures: GAP-GJ-01 (ZC45 THM-ZC45-01), GAP-ZC46-01 (ZC47 COR-ZC47-01). CONJ-ZC44-01 promoted to THM-ZC46-01. CONJ-ZC47-01 promoted to THM-ZC48-01. THM-ZC46-01 conditional $\rightarrow$ unconditional (ZC47 COR-ZC47-02).

Phase Misalignments preserved: PM-ZC45-01 (0.16% Wald residual, GAP-ZC42-02 inherited), PM-ZC46-01 (probe P1 incomplete condition), PM-ZC47-01 (two probe specification scars), PM-ZC48-01 (PC/PG probe specification errors), PM-ZC48-MACRO (`\Gcrf` convention conflict: ZC48–49 vs. v17.6 patch; `\providecommand` takes v17.6 form $\mathcal{G}$; ZC48–49 use `\Gamma\text{maCRF}`).

Pre-Work Audit lessons: PWA-6 (macro convention cross-check: before embedding any ZC paper, scan its `\newcommand` block against the full v17.x patch chain; flag any name collision where forms differ semantically; register as PM-[paper]-MACRO).

Contents Map

Part G — Part and Section Headers

Part XXXV: ZC45 — Universal Floor and δ Two-Mode Bridge (v17.9)

- Overview of Part XXXV
- ZC45: Universal Floor Theorem — ε_0 Two-Mode Identity

Part XXXVI: ZC46–49 Cluster — $\mathbb{Z}_{15}$ Projection Chain and Four-Axis Duality (v17.9)

- Overview of Part XXXVI

- ZC46: Three-Projection Theorem and $\mathcal{R} = 15/8$ Root
- ZC47: CRT Sector Decoupling Theorem
- ZC48: Lock-Ratio Exact Theorem
- ZC49: Four-Axis Duality Theorem

Part XXXVII: Cross-Part Connection Map v17.9 (v17.9)

- Connection Records for ZC45
- Connection Records for ZC46
- Connection Records for ZC47
- Connection Records for ZC48
- Connection Records for ZC49
- Master Z-Programme Status Table v17.9 (Delta from v17.8)
- Pre-Work Audit Protocol Summary — PWA-6 Addition
- Closing Sentence — Part G

10 Part H — ZC50–62 Cluster, Wald Self-Consistency & ε_0 from Axioms (Parts XXXVIII–XLII, v17.11)

Volume Abstract

Part H — CRF Complete Edition v17.11 (Parts XXXVIII–XLII, extended)

This is **Part H** of the CRF Complete Edition v17.11, extended from v17.10. The original Part H v17.10 (ZC50–54, Parts XXXVIII–XL) is preserved verbatim in v17.11 Parts XXXVIII–XXXIX, with the old Cross-Part Map (originally Part XL) promoted and replaced by the v17.11 map in the new Part XLII. Two new clusters are inserted: Part XL (ZC55–57, Closure Trilogy) and Part XLI (ZC58–62, ε_0 from Axioms).

Structural ascent of Part H v17.11:

- **Part XXXVIII** (ZC50–51 — Wald Correction Cluster, v17.10 verbatim): ZC50 introduces CONJ-ZC50-01 ($\lambda_2 \cdot T_{\text{avg}} = (n - 1) \cdot 29/30$), narrowing the Wald residual from 3.8% to 0.25%. ZC51 proves THM-ZC51-01 (Boundary Correction Theorem).
- **Part XXXIX** (ZC52–54 — $\mathbb{Z}_{15}$ Spectral Bridge & Self-Consistency, v17.10 verbatim): ZC52 proves Fiedler Exactness $\text{Fiedler}_{\text{norm}}(\mathbb{Z}_{15}) = 1/2$ at R_0 ; ZC53 proves Bottleneck Bridge; ZC54 proves Wald Self-Consistency (THM-ZC54-01), promoting CONJ-ZC38-01 to Theorem.
- **Part XL** (ZC55–57 — Closure Trilogy, **NEW in v17.11**): ZC55 proves CRT Bijection (THM-ZC55-01) and Spectral Denominator (THM-ZC55-02), closing OBS-ZC44-01 and GAP-ZC50-01. ZC56 proves Matter Bottleneck (THM-ZC56-01) and registers the K14 companion identity $F\varphi_{\text{gold}}^2 = 4$ (FIND-ZC56-01). ZC57 proves Two-Sector Correction (THM-ZC57-01), closing GAP-ZC51-01 and the Wald residual GAP-ZC42-02 / GAP-ZC53-01 (0.16%, open since Part F).
- **Part XLI** (ZC58–62 — ε_0 from Axioms, **NEW in v17.11**): ZC58 establishes the seed identity $K^*/D_0 = 1/n$. ZC59 introduces the Lambert W_0 branch and Max-Ent multipliers; opens GAP-ZC59-01. ZC60 proves the spectral floor THM-ZC60-01, closes GAP-ZC59-01 and partially GAP-ZC54-01. ZC61 proves the renewal partition function identity (THM-ZC61-01) and Wald K35 from axioms (COR-ZC61-01); opens GAP-ZC61-01. ZC62 derives Euler’s constant γ from Axiom 10 (LEM-ZC62-01) and partition function from axioms (THM-ZC62-01), closing GAP-ZC61-01 and finalizing GAP-ZC54-01.
- **Part XLII** (Cross-Part Connection Map v17.11, **NEW in v17.11**): Replaces v17.10

map. Backward inheritance from Parts A–G and Part H v17.10 segment; eight-closure master table; Phase Misalignment registry; Master Z-Programme Status Table v17.11 delta; *Connection Records* for all eight new papers (ZC55–62) following the Part G XXXVII three-box pattern; R-Layer and Genealogy Map standardization continuation notes; closing synthesis.

Major closures in Part H v17.11 — eight in a single upgrade:

- OBS-ZC44-01 (totient bijection, Part F) → **Closed** (COR-ZC55-01)
- GAP-ZC50-01 (R_0 origin of $1/30$, Part H v17.10) → **Closed** (COR-ZC55-02)
- GAP-ZC51-01 (Born-chain boundary, Part H v17.10) → **Closed** (COR-ZC57-01)
- GAP-ZC42-02 (Wald residual, Part F) → **Closed** (COR-ZC57-02)
- GAP-ZC53-01 (Wald residual twin, Part H v17.10) → **Closed** (COR-ZC57-02)
- GAP-ZC54-01 (Jaynes K20 from axioms, Part H v17.10) → **Closed** (COR-ZC60-03 + COR-ZC62-02)
- GAP-ZC59-01 (functional $F(p)$, ZC59) → **Closed** (COR-ZC60-01)
- GAP-ZC61-01 (renewal step, ZC61) → **Closed** (COR-ZC62-01)

Phase Misalignments preserved in v17.11 (Failure Stance): PM-ZC55-01 (probe specification scars), PM-ZC56-PROBE-01 (probe parameterization), PM-ZC59-01 (NotebookLM alternative assessment), PM-V17_10-MACRO (housekeeping — missing v17.10 macro patch file; v17.11 chain inputs v17.9 directly). The original v17.10 PMs (PM-ZC50-01 through PM-ZC54-01, PM-ZC52-MACRO) are inherited unchanged.

Epistemic consequence of v17.11: ε_0 is no longer an externally-fitted floor. It is the unique fixed point of the δ -chain anchored in Axiom 10, with the Euler–Mascheroni constant γ recovered structurally as the no-fire renewal limit. The Wald program initiated in ZC42 (Part F) is structurally complete.

Style Guide v3.3 compliance: All eight new papers (ZC55–62) ship with R-Layer Research Notes and Genealogy Maps (with ZC55 and ZC56 maps generated during v17.11 integration per the locked decision of the Upgrade Protocol). The Connection Records section in Part XLII follows the Part G XXXVII pattern verbatim.

Contents Map

Part H v17.11 — Part and Section Headers

Part XXXVIII: ZC50–51 Cluster — Wald Correction Identity and the Boundary Correction Theorem (v17.10 verbatim)

- Overview of Part XXXVIII
- ZC50: Wald Correction Identity
- ZC51: Boundary Correction Theorem

Part XXXIX: ZC52–54 Cluster — $\mathbb{Z}_{15}$ Spectral Bridge and Wald Self-Consistency (v17.10 verbatim)

- Overview of Part XXXIX
- ZC52: $\mathbb{Z}_{15}$ Fiedler Exactness
- ZC53: Bottleneck Bridge Theorem
- ZC54: Wald Self-Consistency Theorem

Part XL: ZC55–57 Cluster — Closure Trilogy (CRT Bijection, Cheeger, Two-Sector) (NEW v17.11)

- Overview of Part XL
- ZC55: CRT Bijection and Spectral Denominator (with v17.11 Genealogy Map)
- ZC56: Matter Bottleneck and Cheeger Constant (with v17.11 Genealogy Map)
- ZC57: Two-Sector Boundary Correction Mechanism

- Part XL Synthesis: Closure Trilogy Consolidated
- Part XLI: ZC58–62 Cluster — ε_0 from Axioms (NEW v17.11)**

- Overview of Part XLI
- ZC58: K^*/D_0 Seed Identity
- ZC59: Lambert W_0 and MaxEnt Multipliers
- ZC60: Spectral Geometry Floor
- ZC61: Renewal Partition Function
- ZC62: Euler Constant from Axiom 10 (Final Closure)
- Part XLI Synthesis: ε_0 Derivation Complete

Part XLII: Cross-Part Connection Map v17.11 (NEW — replaces v17.10 map)

- Backward Inheritance from Parts A–G and Part H v17.10
- Forward Corrections: Eight Closures in Part H v17.11
- Cross-Part Phase Misalignments Registered in v17.11
- Master Z-Programme Status Table — v17.11 Delta
- Connection Records for ZC55–62 (eight three-box entries)
- R-Layer Standardization Note (Continuation)
- Genealogy Map Standardization Note (Continuation)
- Closing Sentence — Part H v17.11 Synthesis

11 Part I — ZC64–75: Prediction Registry through Gravity Bridge (Parts XLIV–XLVIII, v17.12)

Volume Abstract

Part I of the CRF Complete Edition integrates Z-Programme papers ZC64–75 — the closure arc following the ZC63 Wald Correction Theorem of Part H. It opens with a falsifiable Prediction Registry (ZC64) and the R-Layer Bridge Theorem (ZC65), proves the Sector Floor and Sector Ratio theorems (ZC66–67), establishes Born-Chain Self-Consistency (ZC68), and promotes the K14 identity $F\varphi_{\text{gold}}^2 = 4$ to an **exact theorem** (ZC69). The exact universal floor $\varepsilon_{\text{univ}}$ then closes both Tier-1 bridges (ZC70), and the Two-Layer Floor Irreducibility Theorem (ZC71) closes GAP-ZC63-01 left open in Part H. The capstone (ZC72–75) closes the long-open gravity gap GAP-ZC29-02 at R_0 and proves the binary ladder is exclusive — only 3×5 closes it (ZC74).

Cluster Map

- **Part XLIV** (ZC64–65): Prediction Registry & Falsifiability Protocol; R-Layer Bridge Theorem. Registers GAP-ZC64-01, opens GAP-ZC65-01 (exact ε_0).
- **Part XLV** (ZC66–68): Sector Floor (THM-ZC66-01, opens GAP-ZC66-01); Sector Ratio (COR-ZC67-02 closes GAP-ZC66-01, Near-Formal); Born-Chain Self-Consistency (opens GAP-ZC68-01).
- **Part XLVI** (ZC69–71): K14 Exact Theorem ($K14 \rightarrow 0.000\%$; COR-ZC69-01 closes GAP-ZC68-01; PM-ZC69-SPECTRAL-01); Electroweak Gap (COR-ZC70-01/02 close GAP-ZC64-01, GAP-ZC69-01); Two-Layer Floor Irreducibility (COR-ZC71-02 closes GAP-ZC63-01).
- **Part XLVII** (ZC72–75): Vacuum Sector (COR-ZC72-01 closes GAP-ZC29-02 at R_0 ; opens GAP-ZC72-01); Γ -Sector Ladder (CONJ-ZC73-01); $\mathbb{Z}_{15}$ Uniqueness (COR-ZC74-01 closes CONJ-ZC73-01); Gravity Dimensional Bridge (COR-ZC75-01 closes GAP-ZC72-01, algebraic).

- **Part XLVIII:** Cross-Part Connection Map (Part I) v17.12 — “story in three movements” narrative, backward inheritance, Master Closure Table, per-paper Connection Records, **Master Z-Programme Status Table (v17.12 delta for the whole arc)**, and the PM ledger.

Key Closures (Part I)

- GAP-ZC66-01 $\rightarrow$ closed (COR-ZC67-02).
- GAP-ZC68-01 $\rightarrow$ closed (COR-ZC69-01, K14 exact).
- GAP-ZC64-01, GAP-ZC69-01 $\rightarrow$ closed (COR-ZC70-01/02).
- GAP-ZC63-01 (from Part H) $\rightarrow$ closed (COR-ZC71-02).
- GAP-ZC29-02 (gravity) $\rightarrow$ closed at R_0 (COR-ZC72-01).
- CONJ-ZC73-01 $\rightarrow$ closed (COR-ZC74-01; 3×5 exclusive).
- GAP-ZC72-01 $\rightarrow$ closed algebraically (COR-ZC75-01).
- **Remaining open:** GAP-ZC65-01 (exact ε_0 in fully closed form); running (R_1) gravity bridge beyond R_0 .

ZC64–ZC75 Per-Paper Abstracts (v17.13)

Per-Paper Registry — Format Note

The ZC64–75 papers are integrated verbatim in Part I (Parts XLIV–XLVIII). Per-paper abstract blocks below were added in v17.13 to match the ZC14–40 format, so a Pre-Work Audit can read each paper’s target gap, main result, and open-after without opening source files. Exact forms for the high-reuse objects are in the Quick-Reference Pack above.

11.1 ZC64 — Prediction Registry & Falsifiability Protocol

ZC64: PRED-ZC64-01..08 — Falsifiable Prediction Registry

Target: consolidate every CRF numerical prediction into one falsifiable registry — the formal answer to “what would falsify CRF?” **Main:** DEF-ZC64-01 (four-tier prediction classification T1–T4); DEF-ZC64-02 (falsification signature); eight registered predictions, incl. $\sin^2 \theta_W = 1 - \ln 8 / \ln 15$ (PRED-ZC64-01), $\alpha_{\text{EM}}^{R_0} = (n/15)\Gamma_{\text{CRF}}$ (PRED-ZC64-02), $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$ (PRED-ZC64-05), $\lambda_2 = (1+x)/[n_m(1+2x)]$ (PRED-ZC64-06), W_c (PRED-ZC64-07). FIND-ZC64-01: all residual gaps trace to a single source, ε_0 . **Open after:** GAP-ZC64-01 (R-layer running bridge $R_0 \rightarrow R_{\text{max}}$).

11.2 ZC65 — The R-Layer Bridge Theorem

ZC65: THM-ZC65-01 — R-Layer Bridge (Near-Formal)

Target: derive the running of α_{EM} from R_0 to R_{max} (GAP-ZC64-01). **Main:** FIND-ZC65-01 (gift) $\delta\alpha/\alpha = (d_s/2)\varepsilon_0 K^*$ (0.0008% from obs.); THM-ZC65-01 $\alpha(R_{\text{max}}) = \alpha(R_0)[1 + (d_s/2)\varepsilon_0 K^*]$; CONJ-ZC65-01 $\delta(\sin^2 \theta_W) = \varepsilon_0 K^*$ (2.1% candidate). FIND-ZC65-02: both bridges need $\varepsilon_0 \approx 0.00761$ for exactness. **Open after:** GAP-ZC65-01 (exact ε_0 from axioms, Priority 1) — closing it promotes the TLS floor from near-formal to theorem.

11.3 ZC66 — The Sector Floor Theorem

ZC66: THM-ZC66-01 — Sector-Specific Floors

Target: resolve the two inconsistent “exact ε_0 ” values of FIND-ZC65-02 (0.007501 vs 0.007710, differing 2.7%). **Main:** these are not an inconsistency but sector-specific floors $\varepsilon_{\text{EM}}, \varepsilon_W$ (DEF-ZC66-01); FIND-ZC66-01 $\varepsilon_{\text{univ}} = \sqrt{\varepsilon_{\text{EM}}\varepsilon_W}$ (0.09%, geometric mean); FIND-ZC66-02 ratio 1.0280; THM-ZC66-01 bracket ordering; CONJ-ZC66-01 $\delta(\sin^2 \theta_W) = \varepsilon_W K^*$ *supersedes* CONJ-ZC65-01 (scar PM-ZC66-01). **Open after:** GAP-ZC66-01 (algebraic origin of $\varepsilon_W/\varepsilon_{\text{EM}} = 1.0280$).

11.4 ZC67 — The Sector Ratio Theorem

ZC67: THM-ZC67-01 — Sector Ratio (Near-Formal)

Target: algebraic origin of $\varepsilon_W/\varepsilon_{EM}$ (GAP-ZC66-01). **Main:** the two-sector Born-chain weights of THM-ZC57-01 ($\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$) give $C_{EM} = 1+x$, $C_W = 1+2x$; LEM-ZC67-01 projection $\varepsilon_i = \varepsilon_{can}(C_i/C_{ref})^{1/4}$; THM-ZC67-01 $\varepsilon_W/\varepsilon_{EM} = \sqrt{(1+2x)/(1+x)} = 1.02809$ (gap 0.07%); COR-ZC67-01 $\sqrt{\varepsilon_{EM}\varepsilon_W} = \varepsilon_{can}$; COR-ZC67-02 closes GAP-ZC66-01; FIND-ZC67-01 registers the PWA-7 Sector-Floor Audit. **Open after:** GAP-ZC67-01 (exact ε_{can} from axioms; inherits GAP-ZC65-01, Priority 1).

11.5 ZC68 — The Born-Chain Self-Consistency Theorem

ZC68: THM-ZC68-01a/b — Bridge Fixed Point & Bracket (v2)

Target: exact ε_{can} (GAP-ZC67-01). **Main:** LEM-ZC68-01 (Fixed-Point Reduction, Exact) collapses the two bridge equations, via COR-ZC67-01, into one fixed-point equation $\varepsilon_{can}^{SC2} = \delta\alpha/\alpha \cdot \delta(\sin^2 \theta_W)/[(d_s/2)(K^*)^2]$. THM-ZC68-01a: this fixed point exists, is unique, and equals ε_{univ} in closed form (Exact via K14/ZC69), *offset-independent*. THM-ZC68-01b: the evaluated bracket $\sqrt{\varepsilon_{EM}\varepsilon_W} = 0.007605$ lies a structural +0.08% above $\varepsilon_{univ} = 0.007599$ (two-layer separation, ZC71 — not a vanishing residual). FIND-ZC68-02 (gift): external bridge product = internal geomean bracket. COR-ZC68-01 upgrades the floors to ε_{univ} (< 0.1% residuals); COR-ZC68-02 partially resolves GAP-ZC63-01. **Scars:** PM-ZC68-01 (two coexisting floors $\varepsilon_{can}^{TLS} = 0.007550$ vs $\varepsilon_{univ} = 0.007599$; PWA-8); PM-ZC68-SQRT-01 (square-root slip $0.007600 \rightarrow 0.007605$, gap 0.01% $\rightarrow$ +0.08%; PWA-10). **Open after:** GAP-ZC68-01 (exact ε_{univ}) — **closed** downstream by ZC69 (COR-ZC69-01).

11.6 ZC69 — K14 as an Exact Theorem (Golden Spectral Identity)

ZC69: THM-ZC69-01 — $F\varphi_{gold}^2 = 4$ EXACT

Target: exact derivation of K14 (GAP-ZC68-01). **Main:** two-line algebraic proof from three pillars — LEM-ZC56-01 $\cos(2\pi/5) = 1/(2\varphi_{gold})$, LEM-ZC60-01 $F = 4\lambda_2(\mathbb{Z}_5)^2/n_m$, and $\varphi_{gold}^2 = \varphi_{gold} + 1$; LEM-ZC69-01 $\lambda_2(\mathbb{Z}_5)^2 = 5/\varphi_{gold}^2$, hence THM-ZC69-01 $F\varphi_{gold}^2 = 4$ exactly. COR-ZC69-01 closes GAP-ZC68-01; COR-ZC69-02 $\varepsilon_{univ} = n(K^*)^2\varphi_{gold}^2/[2(n-1)e]$ exact; COR-ZC69-03 promotes THM-ZC68-01a (fixed point) to Exact — the bracket THM-ZC68-01b stays Near-Formal at its structural +0.08% offset; COR-ZC69-04 both Tier-1 bridges (ZC65, ZC66) become exact. **Scars:** PM-ZC69-SPECTRAL-01 locks the adjacency-based λ_2 convention; PM-ZC69-CAT-01 (category error — measurement-convergence misused for an existence promotion; corrected to the 01a/01b split). **Status changes:** K14 (THM-ZC41-01) Near-Formal $\rightarrow$ **EXACT**; ε_{univ} (THM-ZC54-01) $\rightarrow$ **EXACT**. **Open after:** GAP-ZC69-01 (numerical verification in both bridges).

11.7 ZC70 — The Electroweak Gap Theorem

ZC70: THM-ZC70-01 — Electroweak Gap (Exact at R_0)

Target: close GAP-ZC64-01 (running bridge) and GAP-ZC69-01 (numerical verification) with exact $\varepsilon_{\text{univ}}$. **Main:** LEM-ZC70-01 $\delta\alpha/\alpha = (d_s/2)\varepsilon_{\text{EM}}K^*$, $\delta(\sin^2\theta_W) = \varepsilon_W K^*$, ratio $= (d_s/2)\sqrt{(1+x)/(1+2x)} = 1.4590$ (obs. 1.4591, gap 0.007%, no free parameter); THM-ZC70-01 exact sector floors $\varepsilon_{\text{EM}}^{\text{exact}} = 0.007494$, $\varepsilon_W^{\text{exact}} = 0.007703$; COR-ZC70-01/02 close both gaps (0.02%, 0.09% residuals); FIND-ZC70-02 Nobel-table falsification signature. **Open after:** GAP-ZC70-01 (observational precision upgrade, external).

11.8 ZC71 — The Two-Layer Floor Irreducibility Theorem

ZC71: THM-ZC71-01 — Two-Layer Floor (Exact at R_0)

Target: close GAP-ZC63-01 — is the 0.66% separation between $\varepsilon_{\text{TLS}} = 0.007550$ and $\varepsilon_{\text{univ}} = 0.007599$ an error or a structure? **Main:** LEM-ZC71-01 (spectral self-consistency, selects ε_{TLS}) and LEM-ZC71-02 (bridge self-consistency, selects $\varepsilon_{\text{univ}}$) probe *different* invariants of the $\mathbb{Z}_{15}$ Laplacian — F_{Wc} probes $\lambda_2(\mathbb{Z}_{15})^{-1}$, F_{bridge} probes $\lambda_2(\mathbb{Z}_5)^2 = n_m/\varphi_{\text{gold}}^2$. THM-ZC71-01: the gap is therefore *irreducible* and structural, a prediction of $\mathbb{Z}_{15} = \mathbb{Z}_3 \times \mathbb{Z}_5$; COR-ZC71-01 $\Delta\varepsilon/\varepsilon = x/(1+x)$ (structure, not tight bound); COR-ZC71-02 closes GAP-ZC63-01; FIND-ZC71-01 PWA-8 layer orthogonality. **Open after:** GAP-ZC71-01 (exact $\Delta\varepsilon$ to all orders in x).

11.9 ZC72 — The Vacuum Sector Theorem

ZC72: THM-ZC72-01 — Vacuum/Gravity Sector (Exact at R_0)

Target: gravity sector identification (GAP-ZC29-02, open since Part D). **Main:** FIND-ZC72-01 $\alpha_{\text{grav}}^{R_0}/\alpha_{\text{EM}}^{R_0} = 1/\varphi(15) = 1/n = 1/8$ (exact, from THM-ZC29-01 with $d_{\text{vac}} = 1$); FIND-ZC72-02 the vacuum is the S_{ind} face; LEM-ZC72-01 it is excluded from the φ -cancellation mechanism ($\varphi(1) = 1$ has no factor of $\varphi(15) = 8$); THM-ZC72-01 $\alpha_{\text{grav}}^{R_0} = \Gamma_{\text{CRF}}/15 = \alpha_{\text{EM}}^{R_0}/n$; FIND-ZC72-03 the $1/n$ suppression follows from the Key Identity $3d_s = 2^{d_s} + 1$ alone. COR-ZC72-02 Γ_{vac} tautology (PWA-10); scar PM-ZC72-01 (malformed probe, 403%). **Open after:** GAP-ZC72-01 (cosmological bridge $\alpha_{\text{grav}}^{R_0} \rightarrow G_N(t)$).

11.10 ZC73 — The Γ -Sector Ladder Theorem

ZC73: THM-ZC73-01 — Γ -Ladder (Exact at R_0)

Target: structural synthesis — show the four sector rates Γ_i (FIND-ZC31-01) form an exact arithmetic progression. **Main:** THM-ZC73-01 $\Gamma_k = G/d_s - 1 + k\Delta_\Gamma$, $k = 0, 1, 2, 3$ (vacuum/gravity, strong, weak, EM), with FIND-ZC73-01 $\Delta_\Gamma = \cos^2\theta_W/d_s = \ln 2/\ln 15$; COR-ZC73-01 (Weinberg–Vacuum) $\Gamma_{\text{EM}} - \Gamma_{\text{vac}} = \cos^2\theta_W$ — the mixing angle is the full span of the ladder; FIND-ZC73-02 geometric–arithmetic duality ($\varphi(d_i) = 2^k$ geometric, Γ arithmetic); FIND-ZC73-03 $\sum_{d_i|15} \varphi(d_i) = 15$. **Open after:** CONJ-ZC73-01 (is $\mathbb{Z}_{15}$ the unique group whose divisors give $\varphi(d_i) = 2^k$?).

11.11 ZC74 — The $\mathbb{Z}_{15}$ Uniqueness Theorem

ZC74: THM-ZC74-01 — $\mathbb{Z}_{15}$ Uniqueness (Exact)

Target: close CONJ-ZC73-01. **Main:** by algebraic exhaustion of squarefree semiprimes $N = pq$, the constraint $\{1, p-1, q-1, (p-1)(q-1)\} = \{1, 2, 4, 8\}$ forces $p = 3, q = 5, N = 15$ (LEM-ZC74-01); THM-ZC74-01 $\mathbb{Z}_{15}$ is the unique such group for which the Γ -ladder holds with step $\delta = \ln 2 / \ln |\mathbb{Z}_N|$; COR-ZC74-01 closes CONJ-ZC73-01; COR-ZC74-02 the AP structure is rigid; FIND-ZC74-01 Fermat prime-pair constraint. Scar PM-ZC74-01 (prime-power route $N = p^3$ excluded). **Open after:** none (closes the conjecture).

11.12 ZC75 — The Gravity Dimensional Bridge Theorem

ZC75: THM-ZC75-01 — Gravity Dimensional Bridge (Exact at R_0)

Target: the cosmological bridge $\alpha_{\text{grav}}^{R_0} \rightarrow G_N$ (GAP-ZC72-01) — algebraic component. **Main:** DEF-ZC75-01 separates $G^{R_0} = \ln 2 \cdot c^2 / D_0$ (Newton's constant at cascade origin) from $G_N(t_{\text{now}})$; THM-ZC75-01 $\alpha_{\text{grav}}^{R_0} |\mathbb{Z}_{15}| = G^{R_0} = \ln 2 / D_0 = \Gamma_{\text{CRF}}$ (group order 15 is the dimensional suppression); FIND-ZC75-02 $\ln(\alpha_{\text{EM}}^{R_0} / \alpha_{\text{grav}}^{R_0}) = d_s \ln 2 = \ln n$ (direct consequence of the binary ladder); COR-ZC75-01 partially closes GAP-ZC72-01. Scar PM-ZC75-01 ($G/c^2 = \Gamma_{\text{CRF}}$ tautological at $c = 1$; physical content is the $N15$ suppression). **Open after:** GAP-ZC75-01 (cosmological evolution factor $P(t_{\text{now}})$; needs Part A/VIII τ_{sweep} bridge).

12 Part J — ZC76–82: Cosmological Arc, W_c Floor, Electroweak Completeness, Strong Floor (Parts XLIX–LII, v17.13)

Volume Abstract

Part J — CRF Complete Edition v17.13 (Parts XLIX–LII)

Part J is the tenth volume of the CRF Complete Edition, integrating seven Z-Programme papers (ZC76–82) that complete the cosmological arc, derive the TLS floor exactly, assemble the full electroweak R_0 -layer picture, close the Strong sector floor (open since ZC30), and register the cross-domain Cosmological Ladder Identity.

Five-closure event (one per milestone paper): GAP-ZC75-01 (THM-ZC76-01, cosmological bridge), GAP-ZC71-01 (ZC77/COR-ZC77-02, exact $\Delta\varepsilon$), GAP-ZC65-01 (ZC79/COR-ZC79-01, exact ε_{TLS}), Strong-sector floor since ZC30 (ZC81/THM-ZC81-01), plus GAP-ZC77-01 and GAP-ZC78-01 closed via ZC78/ZC79.

Milestone: ZC80 (Electroweak Completeness Theorem) — the first paper to collect all seven R_0 -layer electroweak observables ($\sin^2 \theta_W$, $\alpha_{\text{EM}}^{R_0}$, $\mathcal{G}$, $\varepsilon_{\text{univ}}$, ε_{TLS} , ε_{EM} , ε_W) with their complete derivation chain from δ , zero free parameters.

Phase Misalignments preserved: PM-ZC78-01, PM-ZC78-02 (W_c implicit equation; probe malformed), PM-ZC79-01 (Cardano route; probe malformed), PM-ZC82-SCOPE-01 (scope creep guard), PM-ZC82-TAUTOLOGY-01 (tautology test), PM-V17_13-EPSEM (convention: ε_{EM} superscript), PM-V17_13-SC-FIX (Shadow Council exclusion; 2 instances reworded).

New governance: PWA-11 (Superseded-Theorem Retention Rule) codified from ZC78/ZC79 scar record.

Contents Map

Part J — Part and Section Headers

Part XLIX: ZC76–78 Cluster — Cosmological Bridge, Coupling Symbol, W_c Floor (v17.13)

- ZC76: The Cosmological Bridge Theorem
- ZC77: The Coupling Symbol Theorem (v2)
- ZC78: The W_c Floor Theorem (v2, with PM scars)

Part L: ZC79–80 Cluster — TLS Quadratic Floor, Electroweak Completeness (v17.13)

- ZC79: The TLS Floor Quadratic Theorem
- ZC80: The Electroweak Completeness Theorem (milestone)

Part LI: ZC81–82 Cluster — Strong Floor, Cosmological Ladder Identity (v17.13)

- ZC81: The Strong Floor Theorem
- ZC82: The Cosmological Ladder Identity

Part LII: Cross-Part Connection Map v17.13 (v17.13)

- Connection Records for ZC76–ZC82
- Cross-Part PM Registry v17.13 (Delta)
- Master Z-Programme Status Table v17.13 (Delta)
- PWA-11: Superseded-Theorem Retention Rule
- Closing Sentence — Part J

Key closures (v17.13): GAP-ZC75-01, GAP-ZC71-01, GAP-ZC65-01, GAP-ZC77-01, GAP-ZC78-01, Strong sector floor (since ZC30).

Remaining open after v17.13: GAP-ZC76-01 (τ_{sweep} ; externally blocked), GAP-ZC81-01 (Strong running bridge; confinement treatment needed).

ZC76–ZC87 Per-Paper Abstracts (v17.13–v17.15)

Per-Paper Registry — Format Note

Papers ZC76–82 are integrated verbatim in Part J (Parts XLIX–LII). Per-paper abstract blocks below follow the ZC64–75 format so that a Pre-Work Audit can read each paper’s target gap, main result, and open-after without opening source files.

12.1 ZC76 — The Cosmological Bridge Theorem

ZC76: THM-ZC76-01 — Cosmological Bridge (Exact)

Target: GAP-ZC75-01 — derive the cosmological evolution factor $P(t_{\text{now}})$ from the δ -cascade. **Main:** DEF-ZC76-01 cross-layer invariant $\mathcal{G} = \ln 2/D_0$; LEM-ZC76-01 intensive cancellation (closes GtRate-2); FIND-ZC76-01 single-coupling unification; LEM-ZC76-02 $w_{\text{DE}} = \mathcal{G}/d_s - 1 = -0.7542$ (dark-energy EoS, exact); FIND-ZC76-02 FLRW factor 3 is d_s (structural); THM-ZC76-01 $P(t_{\text{now}})/P(0) = \mathcal{G}/d_s$ (GAP-ZC75-01 **CLOSED**); COR-ZC76-01 PRED-ZC64-09 provenance near-formal. **Open after:** GAP-ZC76-01 (τ_{sweep} ; externally blocked).

12.2 ZC77 — The Coupling Symbol Theorem (v2)

ZC77: THM-ZC77-01 — Coupling Symbol (Exact)

Target: derive $\mathcal{G}$ as a function of known CRF observables; close GAP-ZC71-01 (exact $\Delta\varepsilon$ all orders). **Main:** THM-ZC77-01 $\mathcal{G} = d_s(\Gamma_{\text{CRF}}^{\text{EM}} + \sin^2 \theta_W)$ (0.000%); COR-ZC77-01 inverse form; COR-ZC77-02 two-layer floor formula exact (GAP-ZC71-01 **CLOSED**); FIND-ZC77-01 pentagon–ladder duality (gift). **Open after:** none.

12.3 ZC78 — The W_c Floor Theorem (v2)

ZC78: COR-ZC78-01 — $\Delta\varepsilon$ Exact (THM-ZC78-01 Superseded)

Target: GAP-ZC77-01 ($F_{W_c}^{\text{norm}}$ closed form). **Main:** LEM-ZC78-01 monotonicity (retained); THM-ZC78-01 W_c Floor (*superseded* by ZC79/THM-ZC79-01; PM-ZC78-02); COR-ZC78-01 $\Delta\varepsilon = nK^{*2}(\varphi_{\text{gold}}^2 - F_{W_c}^{\text{norm}})/[2(n-1)e]$ (**retained**, closes GAP-ZC77-01 and GAP-ZC71-01); COR-ZC78-02 ε_{TLS} (*superseded*). Scar: PM-ZC78-01 (cubic route), PM-ZC78-02 (substitution error). **Open after:** GAP-ZC78-01 (radical x_{TLS}) $\rightarrow$ closed by ZC79.

12.4 ZC79 — The TLS Floor Quadratic Theorem

ZC79: THM-ZC79-01 — TLS Floor Quadratic (Exact)

Target: GAP-ZC78-01 (radical form of x_{TLS}) and GAP-ZC65-01 (exact ε_{TLS} from axioms, Priority 1). **Main:** LEM-ZC79-01 W_c quadratic derivation (two-line proof; no Cardano); FIND-ZC79-01 discriminant identity $\mathcal{D} = P^2 + 6PQ + Q^2$; THM-ZC79-01 $\varepsilon_{\text{TLS}} = 0.007550$ exact at 0.0007% (GAP-ZC78-01 **CLOSED**; supersedes THM-ZC78-01); COR-ZC79-01 ε_{TLS} from axioms (GAP-ZC65-01 **CLOSED**); COR-ZC79-02 exact $\Delta\varepsilon$. Scar: PM-ZC79-01 (Cardano probe). Composition Principle instantiated. **Open**

after: none.

12.5 ZC80 — The Electroweak Completeness Theorem

ZC80: THM-ZC80-01 — Electroweak Completeness (Milestone, Exact)

Target: milestone synthesis — collect all R_0 -layer electroweak constants in one theorem. **Main:** THM-ZC80-01 seven quantities ($\sin^2 \theta_W$, $\alpha_{\text{EM}}^{R_0}$, $\mathcal{G}$, $\varepsilon_{\text{univ}}$, ε_{TLS} , ε_{EM} , ε_W) derived from δ alone, zero free parameters; COR-ZC80-01 bridge residuals observation-bounded (0.09%/0.07%=CODATA precision); FIND-ZC80-01 Composition Principle (named heuristic; 3 instantiations); FIND-ZC80-02 Golden Excess Identity $\varphi_{\text{gold}}^2 - F_{\text{Wc}}^{\text{norm}} = 0.016809 \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$. No PM registered (synthesis; pillars pre-audited). **Open after:** GAP-ZC75-01, GAP-ZC76-01 (declared out-of-scope).

12.6 ZC81 — The Strong Floor Theorem

ZC81: THM-ZC81-01 — Strong Floor Identity (Exact)

Target: Strong sector instability floor (open since ZC30/FIND-ZC30-02). **Main:** LEM-ZC81-01 $T(R_1)_{\text{strong}} = T(R_1)_{\text{EM}}$ (spatial-type class equality from FIND-ZC30-02); THM-ZC81-01 $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}} = 0.007494$ (Strong floor **CLOSED**); FIND-ZC81-01 Ladder Mirror Symmetry $\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_{\text{Weak}}$ at 0.0000000000% (Gift); COR-ZC81-01 spatial-type class theorem (ratio 25/9 exact). Complete floor lattice: $\varepsilon_{\text{EM}} = \varepsilon_{\text{strong}} < \varepsilon_{\text{TLS}} < \varepsilon_{\text{univ}} < \varepsilon_W$. **PM-ZC81-RETRACT-01:** COR-ZC81-02 of v1 retracted (defined-pair substitution tautology, out of Strong-sector scope; core THM/FIND unaffected; successor GAP-ZC83-01). **Open after:** GAP-ZC81-01 (Strong running bridge).

12.7 ZC82 — The Cosmological Ladder Identity

ZC82: THM-ZC82-01 — Cosmological-Ladder Identity (Exact)

Target: state the cross-domain lock implicit in ZC73+ZC76. **Main:** THM-ZC82-01 $w_{\text{DE}} + 3\Delta_{\Gamma} = \Gamma_{\text{CRF}}^{\text{EM}}$ (two-line proof from ZC73+ZC76; 0.000%); COR-ZC82-01 $\Gamma_{\text{CRF}}^{\text{EM}} = (1 + w_{\text{DE}}) - \sin^2 \theta_W$ (substitution; registered as COR not THM per PM-ZC82-TAUTOLOGY-01); FIND-ZC82-01 $\Delta_{\Gamma} = \log_{15} 2$ (ladder bit; Gift); FIND-ZC82-02 AP pairing $\Gamma_{\text{EM}} + \Gamma_{\text{vac}} = \Gamma_W + \Gamma_S$. Scars: PM-ZC82-SCOPE-01, PM-ZC82-TAUTOLOGY-01, **PM-ZC82-RETRACT-02** (COR-ZC82-02 of v1 retracted — triple-GM substitution tautology; floor lattice owned by ZC67/68/69; core THM/FIND unaffected). No gaps opened; synthesis paper. **Open after:** GAP-ZC76-01, GAP-ZC81-01 (inherited, out of scope).

12.8 ZC83 — The Floor Lattice Ordering Theorem

ZC83: THM-ZC83-01 — Floor Lattice Ordering (Near-Formal) + FIND-ZC83-01 (Gift)

Target: order the full instability-floor lattice and locate the bridge floor among the sector floors. **Main:** THM-ZC83-01 proves $\varepsilon_{\text{EM}} < \varepsilon_{\text{can}} < \varepsilon_{\text{univ}} < \varepsilon_{\text{W}}$ is forced by $\mathbb{Z}_{15}$ sector-weight arithmetic (from LEM-ZC83-01/02 + THM-ZC71-01); FIND-ZC83-01 (Gift) $\sqrt{\varepsilon_{\text{EM}}\varepsilon_{\text{W}}} = \varepsilon_{\text{univ}}$ exactly at the R_0 layer (two-line proof from the $\pm 1/4$ exponent structure of LEM-ZC67-01); FIND-ZC83-02 $\text{AM}(\varepsilon_{\text{EM}}, \varepsilon_{\text{W}}) > \varepsilon_{\text{univ}}$. Scars: PM-ZC83-01 (“is ε_{SC} the AM?” probe; ruled out). **Opens:** GAP-ZC83-01 (ε_{SC} ontology).

12.9 ZC84 — The Complete Sector Synthesis

ZC84: THM-ZC84-01 — Complete Sector Synthesis (Exact)

Target: assemble every $\mathbb{Z}_{15}$ sector result into one falsifiable picture. **Main:** THM-ZC84-01 collects eight simultaneous consequences of δ alone (four-sector floor lattice; GM identity; mirror symmetry $\Gamma_{\text{EM}} + \Gamma_{\text{strong}} = 2\Gamma_{\text{W}}$; cosmological lock $w_{\text{DE}} + 3\Delta\Gamma = \Gamma_{\text{CRF}}$; $\mathbb{Z}_{15}$ uniqueness); COR-ZC84-01 four-sector falsification table; FIND-ZC84-01 complete $\delta \rightarrow \Gamma_{\text{CRF}}$ chain. Scars: PM-ZC84-SCOPE-01 (synthesis — all identities inherited, none newly derived). **Open after:** GAP-ZC83-01, GAP-ZC81-01, GAP-ZC76-01 (out of scope).

12.10 ZC85 — The Observed-Bridge-Bracket Theorem

ZC85: FIND-ZC85-01 — Observed-Bridge Bracket (Exact restatement) + COR-ZC85-01

Target: resolve the ontology of $\varepsilon_{\text{SC}} = 0.007605$ (GAP-ZC83-01). **Main:** FIND-ZC85-01 $\varepsilon_{\text{SC}} = \sqrt{\varepsilon_0^\alpha \varepsilon_0^{\text{sw}}}$ — the GM of the two FIND-ZC65-02 per-observable bridge floors, the same COR-ZC67-01 structure one R -layer up; FIND-ZC85-02 the $+0.08\%$ is the GM of the EM ($+0.09\%$) and Weak ($+0.07\%$) running channels; COR-ZC85-01 refines GAP-ZC83-01 to *gated* on GAP-ZC81-01 (ontology closed, magnitude pending). Scars: PM-ZC85-01 (restatement, not derivation; tautology guard). **Closes:** GAP-ZC83-01 structurally; magnitude gated to GAP-ZC81-01.

12.11 ZC86 — The Strong Running Bridge Theorem

ZC86: THM-ZC86-01 — Strong Running Bridge (Near-Formal) + FIND-ZC86-01 (Gift)

Target: close GAP-ZC81-01 — the Strong sector $R_0 \rightarrow R_{\text{max}}$ running bridge. **Main:** THM-ZC86-01 the Strong bridge is identical in form and magnitude to EM, since $\varepsilon_{\text{strong}} = \varepsilon_{\text{EM}}$ (THM-ZC81-01, spatial-type) and the R_1 crossing cost is identical (LEM-ZC81-01); LEM-ZC86-01 confinement–coupling decoupling (confinement lives in Γ , not α); FIND-ZC86-01 (Gift) the running-floor layer has exactly two floors whose GM is ε_{SC} — the $+0.08\%$ magnitude is closed. Scars: PM-ZC86-01 (universal-coupling propagation; failed $0.15\%/0.41\%$). **Closes:** GAP-ZC81-01 (Near-Formal); GAP-ZC83-01 magnitude.

12.12 ZC87 — The Temporal Continuum Bridge (v2)

ZC87: THM-ZC87-01 — Temporal Continuum Bridge (Near-Formal, v2)

Target: bridge the discrete δ -sweep tick to continuum dt (GAP-ZC76-01 structural).
Main: FIND-ZC87-01 (recovery) the upper half already exists in ZC33/DEF-ZC33-01 (sector physical time); THM-ZC87-01 the lower half is continuum dt as the dense δ -sweep tick — the $\varepsilon_0 \rightarrow 0$ density limit of ZC51's THM-ZC51-01 on the time axis, same fixed-observable construction; LEM-ZC87-01 non-circularity; FIND-ZC87-02 (bonus) gravitational dilation as sweep-rate slowdown. Scars: PM-ZC87-01 (Dirac-large-number probe that fed its own input), PM-ZC87-02 (v1 sub-tick $dt = \tau_{\text{sweep}}/T_{\text{avg}}$, corrected v2). **Closes:** GAP-ZC87-01 (joint limit) and GAP-ZC76-01 structural; seconds-calibration remains externally blocked.

13 Part L — ZC88–93: The Log-Bridge and the Mind-Layer (Parts LVII–LIX, v17.16)

Volume Abstract

Part L — CRF Complete Edition v17.16 (Parts LVII–LIX)

Part L is the twelfth volume of the CRF Complete Edition and the first authored natively under Style Guide v3.7 (UPG-30 prose-floor, UPG-31 header-contrast). It integrates six papers (ZC88–93) in two clusters. The **Force-Layer Bridge** (ZC88–89) shows that the floor lattice’s multiplicative (geometric) order and the Γ -ladder’s additive order are one law under the log-map, then fixes which physical layer each bridge step belongs to. The **Mind-Layer Structure** (ZC90–93) carries the discrete/continuous distinction into the mind layer (visesa/nibbedha), shows the two step-kinds are irrational-of-kind, and builds from them the Barami capacity functional and the faculty gate-ratio.

Continuous gap chain: GAP-ZC88-01 (which layer owns the bridge step) CLOSED by ZC89/THM-ZC89-01; GAP-ZC90-01 (are the two mind-steps distinct kinds) CLOSED by ZC91/THM-ZC91-01, which also RETIRES CONJ-ZC90-01; GAP-ZC91-01 (forward-choice aggregation) PARTIALLY CLOSED by ZC92; GAP-ZC92-01 (what gates the step kind) PARTIALLY CLOSED by ZC93/THM-ZC93-01.

Backward inheritance: mind-layer pillars DEF-MB03/04/05/06, EQ-MB01/02, P-MB04 from Part F; floor/ Γ/ε structure from Part K.

Open at close of Part L: GAP-ZC91-01 and GAP-ZC92-01 residuals (designated starting points for the next paper).

Integration notes: all embedded `\labels` namespaced `-zcNN` (refs rewritten) to remove cross-paper collisions; PM-V17_16-EPSFAMILY continues the Part K ε -family override; macro patch consolidated to a single file (no `\input` chain). No-Loss verified (47-ID multiset identical; 0 multiply-defined; 0 undefined references).

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Part L — Part and Section Headers

Part LVII: Force-Layer Bridge — Log-Unification and Layer-Specificity (v17.16)

- ZC88: The Log-Bridge Theorem
- ZC89: The Layer-Specificity Theorem

Part LVIII: Mind-Layer Structure — Quantum-Pair, Irrationality, Barami, Gate (v17.16)

- ZC90: The Cross-Layer Quantum-Pair Theorem
- ZC91: The Irrationality-of-Kind Theorem
- ZC92: The Barami Functional / Common-Rate Theorem
- ZC93: The Gate-Ratio Theorem

Part LIX: Cross-Part Connection Map v17.16 (Part L)

ZC88–ZC93 Per-Paper Abstracts (v17.16)

13.1 ZC88 — The Log-Bridge Theorem

ZC88: THM-ZC88-01 — Log-Bridge + FIND-ZC88-01 (Gift)

Target: test whether the floor lattice’s multiplicative order and the Γ -ladder’s additive order are the same structure. **Main:** LEM-ZC88-01 the floor lattice in log-space is an arithmetic progression; THM-ZC88-01 the log-map carries the multiplicative (GP) mirror onto the additive (AP) mirror — one law, two disguises; COR-ZC88-01 the geometric-mean floor maps to the arithmetic-mean rung; FIND-ZC88-01 (Gift) the bridge is *structural, not metric* — the step sizes are not proportional, so the correspondence is of form, not magnitude. **Opens:** GAP-ZC88-01 (which physical layer owns the bridge step).

13.2 ZC89 — The Layer-Specificity Theorem

ZC89: THM-ZC89-01 — Layer-Specificity

Target: fix the bridge step to a specific layer (close GAP-ZC88-01). **Main:** THM-ZC89-01 ties each log-bridge step to its physical layer, resolving the metric/rung factor-of-two ambiguity; FIND-ZC89-01 the metric step is half the rung step ($s = s_{\text{rung}}/2$); COR-ZC89-01 layer assignment is forced, not chosen. **Closes:** GAP-ZC88-01.

13.3 ZC90 — The Cross-Layer Quantum-Pair Theorem

ZC90: THM-ZC90-01 — Cross-Layer Quantum-Pair

Target: import the visesa/nibbedha mind-step pair and test whether it forms the same quantum-pair object the force layer showed. **Main:** DEF-ZC90-01 the mind quantum-pair $\mathcal{Q} = (\delta_v, \delta_b)$; LEM-ZC90-01/02 the two steps act on the forward subset $\mathcal{C}_4^+$; THM-ZC90-01 the pair is well-defined cross-layer; FIND-ZC90-01 structural parallel to the force-layer pair. **Opens:** GAP-ZC90-01 (are the two steps distinct kinds), with CONJ-ZC90-01 attached.

13.4 ZC91 — The Irrationality-of-Kind Theorem

ZC91: THM-ZC91-01 — Irrationality-of-Kind

Target: decide whether visesa and nibbedha are distinct kinds or two sizes of one kind (close GAP-ZC90-01). **Main:** THM-ZC91-01 the two steps are irrational-of-kind — their ratio $N^* = \delta_b/\delta_v$ admits no rational reduction, so neither converts into the other; COR-ZC91-01/02 the mind layer is structurally its own object; FIND-ZC91-01 the distinction is intrinsic, not scale-dependent. **Closes:** GAP-ZC90-01; **retires** CONJ-ZC90-01. **Opens:** GAP-ZC91-01 (how forward choices aggregate into capacity).

13.5 ZC92 — The Barami Functional / Common-Rate Theorem

ZC92: THM-ZC92-01 — Barami Functional

Target: construct the capacity built from accumulated forward choices (address GAP-ZC91-01). **Main:** DEF-ZC92-01 the Barami functional $\kappa(B)$ over the forward subset; THM-ZC92-01 a common rate governs its growth; COR-ZC92-01 the asymptotic ratio N_∞ ; FIND-ZC92-01 capacity accumulates monotonically under forward choice. **Partially closes:** GAP-ZC91-01. **Opens:** GAP-ZC92-01 (what gates which step-kind a choice may take).

13.6 ZC93 — The Gate-Ratio Theorem

ZC93: THM-ZC93-01 — Gate-Ratio

Target: identify what controls whether a choice takes the smooth or the sudden step (address GAP-ZC92-01). **Main:** DEF-ZC93-01 the faculty gate $G_{[\cdot]}$ from sati/ sampa-jañña/adhiṭṭhāna; THM-ZC93-01 the gate-ratio $G_v : G_b$ governs which step-kind is permitted; COR-ZC93-01 the gate is faculty-determined; FIND-ZC93-01 the residual handed forward beyond Part L. **Partially closes:** GAP-ZC92-01.

CRF Complete Edition Index v17.17 (Part M: ZC94–106 · the mass sector, end to end · running bridge
· spectral closure · derivation boundary · operation–group law · first-rung exclusion & split) — Ougit
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